JELLYMATIC GUIDE



GAMES + JELLYFIN + POCKET LINK

Jellymatic

Jellyfin on your ModRetro Chromatic

CREATED BY CHAOTICSPARTAN1

Every page of the documentation, printed from the repository's Markdown.
For the interactive version, download the repository and double-click Open-Documentation.html.

CREATED BY CHAOTICSPARTAN1

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START HERE

JELLYMATIC / README.MD



A START-UP MENU FOR THE CHROMATIC THAT KEEPS YOUR GAMES EXACTLY AS THEY ARE, A JELLYFIN APP, A LINK TO THE ANALOGUE POCKET, AND A PC TOOL THAT INSTALLS, CHECKS AND UNDOES EVERYTHING.

READ THE GUIDE: DOCS/JELLYMATIC-GUIDE.PDF (OPENS RIGHT ON GITHUB), OR DOWNLOAD AND DOUBLE-CLICK `Open-Documentation.html` FOR THE INTERACTIVE VERSION.

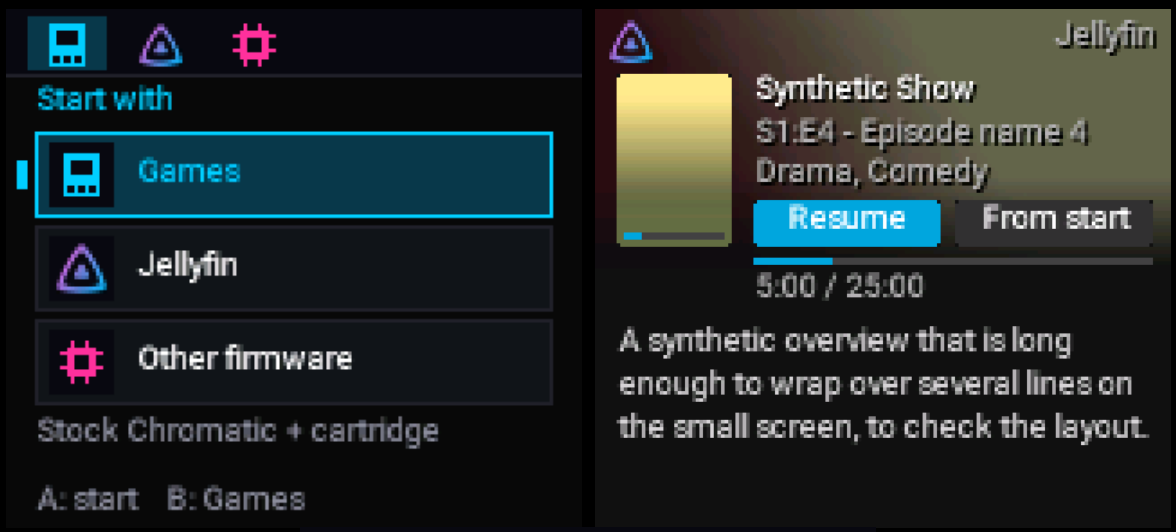
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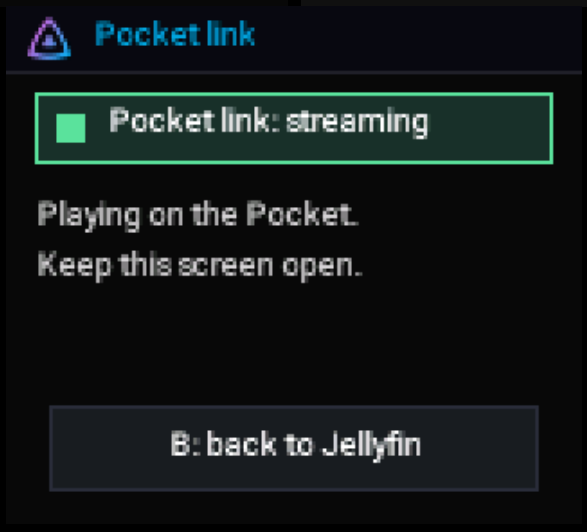
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WHAT IT IS

PART	WHAT IT DOES
START-UP MENU (ESP32 "LAUNCHER")	SHOWS **GAMES**, **JELLYFIN** AND **OTHER FIRMWARE** WHEN YOU SWITCH THE CONSOLE ON. GAMES IS ALWAYS THERE AND STARTS THE CONSOLE'S OWN, UNTOUCHED FIRMWARE.

PART	WHAT IT DOES
JELLYFIN APP	BROWSE YOUR JELLYFIN LIBRARY AND PLAY VIDEO WITH SOUND ON THE CHROMATIC, OVER WI-FI. WI-FI AND THE JELLYFIN SIGN-IN ARE SET UP ON THE CONSOLE ITSELF.
POCKET LINK	AN UPDATED CONSOLE CHIP (FPGA) IMAGE, 18.41, THAT LETS THE CHROMATIC FEED AN ANALOGUE POCKET OVER A LINK CABLE. THE POCKET SIDE IS POCKETFIN.
CHROMATIC MANAGER	A WINDOWS PROGRAM (WINDOW, TEXT MENU, LOCAL API AND MCP SERVER) THAT INSTALLS, UPDATES, CHECKS, BACKS UP AND REMOVES ALL OF THE ABOVE, ONE APPROVE AT A TIME.
REMOTE PILOT (BUNDLED)	SEE AND PRESS THE CONSOLE'S MENUS FROM YOUR PC, IN YOUR BROWSER OR IN THE MANAGER'S OWN TAB. THE FULL TOOL IS REMOTEPILOT.







ALL PICTURES ARE DRAWN BY THE PROJECT'S OWN PREVIEW RENDERERS WITH MADE-UP SAMPLE DATA.

REMOTE PILOT: YOUR CHROMATIC ON YOUR PC SCREEN

REMOTE PILOT LETS YOU **SEE AND CONTROL THE CHROMATIC FROM YOUR PC**: ITS LIVE SCREEN, ITS BUTTONS, ITS LOGS AND SCREENSHOTS, OVER THE USB CABLE OR YOUR HOME WI-FI. A COPY IS BUNDLED IN `remote-pilot/` SO THE MANAGER'S **REMOTE PILOT** TAB WORKS OUT OF THE BOX.

WHY IT IS INCLUDED:

- **EASIER SET-UP AND TROUBLESHOOTING**: READ THE CONSOLE'S MENUS AND MESSAGES ON A BIG SCREEN INSTEAD OF SQUINTING AT THE HANDHELD, AND PRESS ITS BUTTONS WITH THE MOUSE.
- **CLEAN BUG REPORTS**: SAVE SCREENSHOTS AND LOGS WITH ADDRESSES, NETWORK NAMES AND CODES MASKED.
- **AI ASSISTANTS, SAFELY**: CLAUDE, CODEX OR ANOTHER ASSISTANT CAN TEST OR DEMO THE CONSOLE THROUGH REMOTE PILOT'S COMMAND LINE (IT PRESSES BUTTONS AND READS THE SCREEN; IT CANNOT CHANGE FIRMWARE).
- **REMOTE DEMOS**: SHOW THE CONSOLE'S SCREENS TO SOMEONE ELSE FROM YOUR PC.

IT IS **OPTIONAL**, STAYS **ON YOUR OWN PC AND NETWORK** BY DEFAULT, AND IS **NEVER NEEDED** TO WATCH JELLYFIN. FULL DOCUMENTATION: REMOTEPILOT AND DOCS/REMOTE-PILOT.MD.

QUICK START (PLAIN ENGLISH)

1. **READ BEFORE YOU START.** IT EXPLAINS WHY YOUR GAMES KEEP WORKING AND WHAT IS CHANGED ON THE CONSOLE.
2. **GET THE TOOLS** (WHAT YOU NEED): A WINDOWS PC, PYTHON 3.12+, THE USB CABLE, OPENFPGALOADER WITH GWU2X SUPPORT (FOR THE FIRST BACKUP), AND YOUR CHROMATIC SWITCHED **ON**.
3. **OPEN CHROMATIC MANAGER:** DOUBLE-CLICK `Chromatic-Manager-Window.cmd` (FIRST RUN SETS EVERYTHING UP; SEE INSTALL WITH CHROMATIC MANAGER).
4. **PRESS THE ONE BIG BUTTON** THE MANAGER SHOWS (INSTALL), THEN **APPROVE**. THE FIRST TIME IT TAKES A FULL BACKUP OF BOTH CHIPS (ABOUT 15 MINUTES, NOTHING WRITTEN), THEN WRITES THE START-UP MENU, READS EVERY BYTE BACK AND CHECKS THE CONSOLE STARTS. THE BACKUP'S SECOND COPY IS USUALLY ON THE SAME DISK; PUT IT ON ANOTHER DISK OR A USB STICK FOR REAL SAFETY (INSTALL).
5. **ON THE CONSOLE:** SWITCH IT OFF AND ON, CHOOSE **JELLYFIN**, PRESS **SELECT** FOR SETUP, ADD YOUR WI-FI AND SIGN IN TO JELLYFIN WITH **QUICK CONNECT** (FIRST RUN ON THE CONSOLE).
6. **PLAY:** CHOOSE A FILM OR EPISODE AND PRESS A (USING JELLYFIN).

SOMETHING WENT WRONG? RECOVERY EXPLAINS WHAT THE MANAGER DOES BY ITSELF, AND HOW TO PUT EVERYTHING BACK, INCLUDING WITH MODRETRO'S **OFFICIAL** UPDATE TOOL.

DOCUMENTATION (WIKI)

START AT **DOCS/HOME.MD**. PAGES: WHAT YOU NEED · BEFORE YOU START · INSTALL WITH CHROMATIC MANAGER · FIRST RUN ON THE CONSOLE · USING JELLYFIN ON THE CHROMATIC · HOSTING JELLYFIN · POCKET LINK · REMOTE PILOT · RECOVERY · FAQ · BUILDING FROM SOURCE · AI PROMPT EXAMPLES · CREDITS & LICENCES.

AS ONE PDF: DOCS/JELLYMATIC-GUIDE.PDF HOLDS EVERY PAGE, WITH A CLICKABLE TABLE OF CONTENTS; GITHUB SHOWS IT RIGHT IN THE BROWSER.

OFFLINE: DOWNLOAD OR CLONE THIS REPOSITORY, THEN DOUBLE-CLICK `Open-Documentation.html` IN ITS MAIN FOLDER. IT OPENS EVERY PAGE ABOVE IN YOUR BROWSER, WITH PICTURES, WITHOUT A SERVER OR AN INTERNET CONNECTION (KEEP IT IN THE REPOSITORY FOLDER: IT LOADS THE PICTURES FROM `docs/`). IT

IS BUILT FROM THE MARKDOWN PAGES BY `scripts/build_docs_html.py` (THE PDF BY `scripts/build_docs_pdf.py`).

WHAT IS IN THIS REPOSITORY

chromatic/launcher/	start-up menu (ESP-IDF 5.3 app, custom bootloader, host tests)
chromatic/jellyfin/	the console's Jellyfin app (shared C, host-tested)
chromatic/fpga-media/	console chip (FPGA) sound hooks, Verilog + test benches
rtl/	Pocket link bridge and link-port RTL + test benches
fpga/images/	the exact sources and build receipts of the reviewed FPGA images 18.40 and 18.41
scripts/	Chromatic Manager (GUI, text menu, API, MCP), FPGA staging, tests, preview renderer
remote-pilot/	Remote Pilot receiver + web viewer, Chromatic package and presets
pocket/, shared/	the small shared parts the launcher uses (link framing, fonts)
docs/	the wiki

NOT IN THIS REPOSITORY: MODRETRO'S FIRMWARE OR UPDATER, ANY CONSOLE BACKUP, BUILT FIRMWARE IMAGES, THE GOWIN FPGA TOOLS OR LICENCE, AND ANYTHING PERSONAL. THE START-UP MENU YOU CAN BUILD YOURSELF. **JELLYFIN SOUND AND THE POCKET LINK NEED THE OFFICIAL RELEASE FILES** OF THE CONSOLE CHIP (FPGA) IMAGES 18.40 AND 18.41 (`evt1_x2.fs` AND `evt1_x2.bin`), PUBLISHED SEPARATELY AS RELEASE ASSETS: A BUILD OF YOUR OWN WILL NOT MATCH THE SHA-256 PINS THE MANAGER CHECKS, SO IT IS REFUSED (SEE WHAT YOU NEED). YOUR OWN FILES (WI-FI LIST, JELLYFIN SIGN-IN, BACKUPS) LIVE IN A PER-USER FOLDER, `%LOCALAPPDATA%\Jellymatic`, CREATED ON FIRST RUN AND NEVER PART OF GIT.

SIBLING PROJECTS

- POCKETFIN: JELLYFIN ON THE ANALOGUE POCKET (THE POCKET SIDE OF THE POCKET LINK).
- JELLYMATIC: THIS PROJECT.
- REMOTEPILOT: REMOTE PILOT, THE FULL PC REMOTE-CONTROL TOOL.

LICENCE

JELLYMATIC IS FREE SOFTWARE: **GPL-3.0-OR-LATER** (LICENSE), WITH ADDITIONAL TERMS UNDER GPLV3 SECTION 7 (ADDITIONAL-TERMS.MD): KEEP THE AUTHOR ATTRIBUTION "CHAOTICSPARTAN1" (7(B)), MARK MODIFIED VERSIONS AS MODIFIED (7(C)), AND ADDITIONAL PERMISSIONS FOR FPGA BITSTREAMS

THAT COMBINE JELLYMATIC'S RTL WITH MODRETRO'S DESIGN AND GOWIN IP, AND FOR LINKING THE CONSOLE FIRMWARE WITH ESPRESSIF'S BINARY-ONLY WI-FI/PHY LIBRARIES AND THE ESP32 ROM. PARTS MADE BY OTHERS KEEP THEIR OWN LICENCES: SEE NOTICE, THIRD-PARTY-NOTICES.MD AND CREDITS & LICENCES.

CONTRIBUTING AND SECURITY

PULL REQUESTS, BUG FIXES AND BUG REPORTS ARE WELCOME: CONTRIBUTING.MD. TO REPORT A SECURITY PROBLEM PRIVATELY: SECURITY.MD. NEVER POST WI-FI DETAILS, SERVER ADDRESSES, TOKENS OR LOGS WITH ADDRESSES.

CREDITS

CREATED BY **CHAOTICSPARTAN1**. BUILT ON MODRETRO'S OPEN-SOURCE CHROMATIC FPGA SOURCE (GPL-3.0), ESPRESSIF ESP-IDF (APACHE-2.0), THE MISTER CORE ("GAMEBOY_MISTER"), AND THE JELLYFIN LOGO (CC BY-SA 4.0, THE JELLYFIN CONTRIBUTORS). SMALL PARTS COME FROM **SHINY RELAY** BY SHINY GEN (GPL-3.0): ITS `uart_tx_busy` FIX IN THE 18.41 IMAGE AND THE REQUEST-ONCE STATUS CHANNEL PATTERN OF THE POCKET LINK BRIDGE. THANKS ALSO TO THE AUTHORS OF CHROMATIX, WHOSE OPEN DOCUMENTATION WAS READ AS A REFERENCE. FULL LIST: CREDITS & LICENCES.



Jellymatic

Jellymatic wiki

JELLYFIN ON YOUR MODRETRO CHROMATIC, WITH A START-UP MENU THAT KEEPS YOUR GAMES EXACTLY AS THEY WERE.

READ IT AS ONE PDF: JELLYMATIC-GUIDE.PDF (OPENS RIGHT ON GITHUB). **OR OFFLINE:** DOWNLOAD OR CLONE THE REPOSITORY, THEN DOUBLE-CLICK `Open-Documentation.html` IN ITS MAIN FOLDER. EVERY PAGE HERE OPENS IN YOUR BROWSER, WITH PICTURES, WITH NO INTERNET CONNECTION NEEDED.

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PAGE	READ IT WHEN
WHAT YOU NEED	BEFORE ANYTHING ELSE: THE CONSOLE, THE PC, THE CABLE, THE SERVER
BEFORE YOU START	YOU WANT TO KNOW WHY YOUR GAMES KEEP WORKING AND WHAT CHANGES

PAGE	READ IT WHEN
INSTALL WITH CHROMATIC MANAGER	YOU ARE READY TO INSTALL OR UPDATE
FIRST RUN ON THE CONSOLE	WI-FI, JELLYFIN SIGN-IN, POCKET LINK SET-UP ON THE CONSOLE
USING JELLYFIN ON THE CHROMATIC	BROWSING AND PLAYING
HOSTING JELLYFIN	YOUR SERVER IS NOT REACHABLE OVER WI-FI YET
POCKET LINK	YOU HAVE AN ANALOGUE POCKET AND A LINK CABLE
REMOTE PILOT	YOU WANT TO SEE AND PRESS THE CONSOLE'S MENUS FROM THE PC
RECOVERY	SOMETHING WENT WRONG, OR YOU WANT TO UNDO EVERYTHING
FAQ	SHORT ANSWERS
BUILDING FROM SOURCE	YOU WANT TO BUILD THE START-UP MENU, THE FPGA IMAGES OR THE MANAGER YOURSELF
AI PROMPT EXAMPLES	YOU USE CLAUDE, CODEX OR ANOTHER ASSISTANT WITH THE MANAGER
CREDITS & LICENCES	WHO MADE WHAT, AND UNDER WHICH LICENCE
DESIGN NOTES	YOU READ THE CODE AND MEET A REFERENCE TO AN INTERNAL NOTE

IN ONE PICTURE

```
 Switch on ──► start-up menu ──┬─► Games ............ the console's own firmware, untouched
(always there)
                               ├─► Jellyfin ......... browse and play over Wi-Fi (Select:
set-up)
                               │      └─► Pocket link  feed an Analogue Pocket over a link
cable
                               └─► Other firmware ... the stock firmware, listed and
protected
 Hold any button while switching on ──► straight to Games, no menu
```



THE PC SIDE IS **CHROMATIC MANAGER**: ONE WINDOW WITH ONE MAIN BUTTON AT A TIME. IT BACKS UP THE CONSOLE, INSTALLS OR UPDATES THE START-UP MENU, READS EVERY BYTE BACK, AND CAN UNDO ALL OF IT.

WHAT YOU NEED



What you need

CONTENTS: FOR THE START-UP MENU AND JELLYFIN · FOR JELLYFIN SOUND AND THE POCKET LINK · NOT NEEDED

FOR THE START-UP MENU AND JELLYFIN

YOU NEED	WHY	NOTES
A **MODRETRO CHROMATIC** WITH ITS **OFFICIAL FIRMWARE**	THE START-UP MENU SITS NEXT TO IT	TESTED WITH OFFICIAL MCU FIRMWARE V4.6 AND FPGA V18.14. THE MANAGER READS THE CONSOLE FIRST AND **REFUSES** LAYOUTS OR FIRMWARE IT DOES NOT KNOW, RATHER THAN GUESS.
A **CARTRIDGE** IN THE CONSOLE	THE MENU DRAWS OVER A RUNNING CARTRIDGE	ANY WORKING CARTRIDGE OR FLASH CART. WITHOUT ONE, THE CONSOLE SIMPLY STARTS AS NORMAL.
A **WINDOWS 10 OR 11 PC**	CHROMATIC MANAGER RUNS THERE	MACOS/LINUX: THE CODE IS PYTHON AND MOSTLY PORTABLE, BUT ONLY WINDOWS IS TESTED.
PYTHON 3.12 OR NEWER	RUNS CHROMATIC MANAGER FROM SOURCE	FROM PYTHON.ORG. `Setup-Jellymatic.ps1` INSTALLS THE REST.
A **USB-C DATA CABLE**	PC ⇔ CONSOLE	THE CONSOLE MUST BE SWITCHED **ON** WHILE THE MANAGER WORKS.

YOU NEED	WHY	NOTES
A **JELLYFIN SERVER** ON YOUR NETWORK, WITH **QUICK CONNECT** TURNED ON	THE CONSOLE SIGNS IN WITH ITS OWN CODE	SEE HOSTING JELLYFIN.
2.4 GHZ WI-FI	THE CONSOLE'S RADIO	A PHONE HOTSPOT WORKS. HOTEL-STYLE SIGN-IN PAGES ARE NOT SUPPORTED.
THE **START-UP MENU IMAGE**	WHAT THE MANAGER INSTALLS	BUILD IT (BUILDING FROM SOURCE) OR COPY IT FROM A RELEASE INTO `release/launcher/`.
OPENFPGALOADER WITH **GWU2X** CABLE SUPPORT	THE FIRST FULL BACKUP READS THE CONSOLE CHIP (FPGA) TOO, OVER THE CONSOLE'S USB	NOT SHIPPED WITH JELLYMATIC (APACHE-2.0). READY-MADE DOWNLOADS OFTEN LACK GWU2X SUPPORT, SO IT USUALLY HAS TO BE **BUILT**: SEE MODRETRO'S OWN INSTRUCTIONS IN ITS CHROMATIC FPGA SOURCE (README, "FLASHING": OPENFPGALOADER MUST BE COMPILED WITH GWU2X CABLE SUPPORT) AND OPENFPGALOADER'S BUILD GUIDE. PUT IT AT `%LOCALAPPDATA%\Jellymatic\tools\openFPGALoader\openFPGALoader.exe` OR SET `JELLYMATIC_OPENFPGALOADER`.
THE GOWIN **GWU2X USB DRIVER**	THE JTAG BRIDGE OPENFPGALOADER TALKS TO	USUALLY INSTALLED BY MODRETRO'S OFFICIAL UPDATER.

FOR JELLYFIN SOUND AND THE POCKET LINK

THESE CHANGE THE CONSOLE'S **GAME CHIP (FPGA)**. THEY ARE OPTIONAL: THE START-UP MENU AND SILENT VIDEO WORK WITHOUT THEM.

YOU NEED	WHY
A **FULL BACKUP OF YOUR CONSOLE** (BOTH CHIPS)	THE MANAGER WILL ONLY CHANGE THE FPGA WHEN IT CAN PUT YOUR ORIGINAL IMAGE BACK. IT TAKES THIS BACKUP BY ITSELF THE FIRST TIME (INSTALL).
OPENFPGALOADER AND THE **GWU2X DRIVER**	AS ABOVE: THEY READ AND WRITE THE FPGA OVER THE CONSOLE'S USB.
THE **OFFICIAL RELEASE FILES** OF THE FPGA IMAGES 18.40 (SOUND) AND 18.41 (SOUND + POCKET LINK)	THE SOUND AND THE POCKET LINK **NEED THE OFFICIAL RELEASE FILES** (`evt1_x2.fs` AND `evt1_x2.bin` OF EACH IMAGE), PUBLISHED SEPARATELY AS **RELEASE ASSETS** OF THIS PROJECT. PUT EACH PAIR IN ITS FOLDER'S `build\` (`fpga/images/sound-18.40/build/`, `fpga/images/link-18.41/build/`). THIS REPOSITORY HAS THE EXACT SOURCES AND RECEIPTS, BUT NO BINARIES. **A BUILD OF YOUR OWN WILL NOT MATCH THE PINS** (IT IS ALMOST NEVER BYTE-IDENTICAL TO THE REVIEWED IMAGE), AND THE MANAGER REFUSES IT ON PURPOSE: ONLY THE REVIEWED IMAGE WAS TRIED ON A CONSOLE (BUILDING FROM SOURCE).
FOR THE POCKET LINK: AN **ANALOGUE POCKET**, **POCKETFIN**, AND A **LINK CABLE**	SEE POCKET LINK.

NOT NEEDED

- NO DEVELOPER CARTRIDGE, UART ADAPTER OR SOLDERING.
- NO ACCOUNT ANYWHERE, AND NOTHING IS UPLOADED. YOUR WI-FI AND JELLYFIN DETAILS STAY ON YOUR PC (`%LOCALAPPDATA%\Jellymatic`) AND YOUR CONSOLE.

- NO MODRETRO FILES: JELLYMATIC NEVER SHIPS THEIR FIRMWARE OR UPDATER. FOR OFFICIAL UPDATES YOU USE **MODRETRO'S OWN UPDATE TOOL**, DOWNLOADED FROM MODRETRO (RECOVERY).

NEXT: BEFORE YOU START

BEFORE YOU START

JELLYMATIC / DOCS/BEFORE-YOU-START.MD

Before you start: keeping your games working

CONTENTS: HOW THE START-UP MENU WORKS · GAMES IS ALWAYS THERE · WHAT IS CHANGED ON THE CONSOLE · WHAT IS NEVER CHANGED · IF SOMETHING FAILS

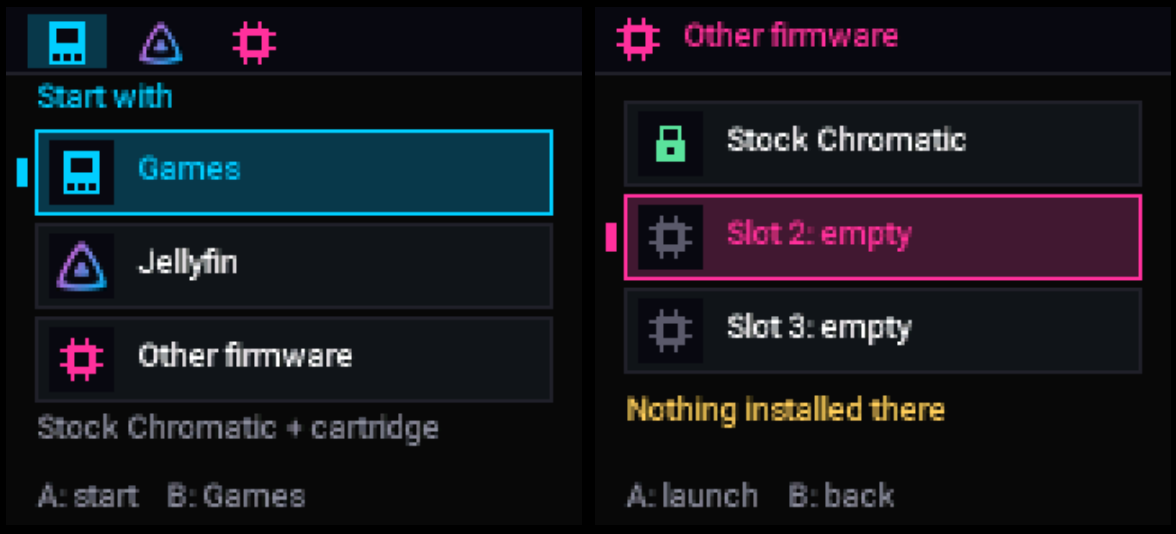
HOW THE START-UP MENU WORKS

THE CHROMATIC HAS TWO CHIPS: AN **ESP32** (MENUS, WI-FI, SETTINGS) AND AN **FPGA** (THE GAME HARDWARE ITSELF). JELLYMATIC ADDS A SMALL **START-UP MENU** TO THE ESP32, **NEXT TO** THE OFFICIAL FIRMWARE, NOT INSTEAD OF IT.

WHEN YOU SWITCH THE CONSOLE ON, A TINY BOOT PROGRAM MAKES ONE DECISION:

HOW THE CONSOLE STARTED	WHAT RUNS
YOU SWITCHED IT ON	THE START-UP MENU (IF ITS IMAGE CHECKS OUT; OTHERWISE THE OFFICIAL FIRMWARE)
ANYTHING ELSE: A RESTART, A CRASH, A WATCHDOG	THE OFFICIAL FIRMWARE, DIRECTLY
YOU HELD ANY BUTTON WHILE SWITCHING ON	THE OFFICIAL FIRMWARE, DIRECTLY, NO MENU

THERE IS NO SETTING, FLAG OR SAVED STATE THAT COULD GO WRONG: THE REASON FOR THE START IS THE ONLY INPUT.



GAMES IS ALWAYS THERE

- **GAMES** (OR **B** ON THE MENU) RESTARTS THE CONSOLE INTO ITS OWN FIRMWARE AND YOUR CARTRIDGE. IT IS THE SAME AS BEFORE JELLYMATIC, WITH YOUR SETTINGS, PALETTES AND SAVES.
- **OTHER FIRMWARE** LISTS THE STOCK CHROMATIC FIRST, MARKED "PROTECTED, ALWAYS HERE".
- THE MENU WAITS FOR YOUR CHOICE; NOTHING STARTS BY ITSELF, AND NOTHING IS LOST BY LEAVING IT OPEN.
- CARTRIDGES, FLASH CARTS, THEIR SD CARDS AND SAVES ARE NEVER READ OR WRITTEN BY JELLYMATIC.

WHAT IS CHANGED ON THE CONSOLE

ONLY THREE AREAS OF THE ESP32'S FLASH:

AREA	WHAT
BOOT PROGRAM (`0x1000`)	A SMALL CUSTOM BOOTLOADER BUILT FROM ESPRESSIF'S ESP-IDF 5.3
PARTITION TABLE (`0x8000`)	THE THREE OFFICIAL ROWS, UNCHANGED AND IN ORDER, PLUS ONE ROW FOR THE MENU
MENU AREA (`0x260000` ONWARD)	THE START-UP MENU AND, AT THE VERY END, YOUR WI-FI AND JELLYFIN SET-UP

OPTIONAL, ONLY IF YOU CHOOSE JELLYFIN SOUND OR THE POCKET LINK: THE FPGA IMAGE IS REPLACED BY A VERSION OF MODRETRO'S OWN OPEN-SOURCE

DESIGN WITH SOUND HOOKS (18.40) OR WITH SOUND HOOKS AND THE POCKET LINK BRIDGE (18.41). IT STILL PLAYS EVERY GAME; IT REPORTS ITSELF WITH THE SAME MAJOR VERSION (18) SO MODRETRO'S TOOLS STILL RECOGNISE THE CONSOLE.

WHAT IS NEVER CHANGED

- THE **OFFICIAL FIRMWARE** SLOT, YOUR **SETTINGS** (NVS) AND **RADIO CALIBRATION** (PHY DATA).
- THE CARTRIDGE, ITS SAVES, AND ANY FLASH CART.

IF SOMETHING FAILS

EVERY CHANGE IS MADE BY CHROMATIC MANAGER, AND EVERY WRITE:

1. STARTS FROM A **BACKUP** OF THE CONSOLE (THE FIRST TIME: BOTH CHIPS, EACH READ TWICE), SAVED WITH A SECOND COPY ON YOUR PC (PUT THAT COPY ON ANOTHER DISK OR A USB STICK IF YOU CAN);
2. IS **WRITTEN DOWN IN A JOURNAL** BEFORE IT HAPPENS, SO AN INTERRUPTED JOB CAN BE FINISHED OR UNDONE;
3. IS **READ BACK** AND COMPARED BYTE FOR BYTE, THEN THE WHOLE FLASH IS CHECKED;
4. NEEDS **YOUR APPROVE** IN THE MANAGER'S WINDOW (AN AI ASSISTANT CAN ASK, BUT YOU DECIDE);
5. ENDS WITH A **START CHECK**: THE CONSOLE MUST START AND KEEP RUNNING.

THE ESP32 ALSO HAS A RECOVERY MODE IN ITS READ-ONLY ROM THAT NO FLASH CONTENT CAN BLOCK, SO THE MANAGER (OR MODRETRO'S OFFICIAL UPDATER) CAN ALWAYS REACH IT AGAIN. SEE RECOVERY.

NEXT: INSTALL WITH CHROMATIC MANAGER

INSTALL WITH CHROMATIC MANAGER

Install with Chromatic Manager

CONTENTS: 1. SET UP THE PC · 2. GET THE START-UP MENU IMAGE · 3. INSTALL · THE WINDOW · WI-FI AND JELLYFIN FROM THE PC · FULL BACKUP BEFORE ANY FPGA CHANGE · JELLYFIN SOUND · UPDATE THE POCKET LINK · TEXT MENU · YOUR FILES

1. SET UP THE PC

1. INSTALL **PYTHON 3.12+** FROM PYTHON.ORG (TICK "ADD PYTHON.EXE TO PATH").
2. DOWNLOAD THIS REPOSITORY (CODE › DOWNLOAD ZIP, OR `git clone`) AND UNZIP IT, FOR EXAMPLE TO `C:\Jellymatic`.
3. RIGHT-CLICK `Setup-Jellymatic.ps1` › **RUN WITH POWERSHELL** (OR `powershell -ExecutionPolicy Bypass -File Setup-Jellymatic.ps1`). IT CREATES YOUR PRIVATE FOLDER `%LOCALAPPDATA%\Jellymatic` AND INSTALLS THE THREE PYTHON PACKAGES THE MANAGER NEEDS. IT NEVER TOUCHES A CONSOLE AND NEVER UPLOADS ANYTHING.

2. GET THE START-UP MENU IMAGE

THE MANAGER INSTALLS THE THREE FILES IN `release\launcher\` (`bootloader\bootloader.bin`, `partition_table\partition-table.bin`, `chromatic_launcher.bin`). EITHER COPY THEM THERE FROM A PUBLISHED RELEASE, OR BUILD THEM YOURSELF WITH `scripts\Build-ChromaticLauncher.ps1` (BUILDING FROM SOURCE).

3. INSTALL

1. PLUG THE CHROMATIC INTO THE PC WITH THE USB CABLE AND **SWITCH IT ON**, WITH A CARTRIDGE IN.
2. DOUBLE-CLICK `Chromatic-Manager-Window.cmd`.
3. THE HOME TAB CHECKS THE CONSOLE (ABOUT 20 SECONDS; THE FIRST TIME, A FULL READ TAKES ABOUT 6 MINUTES), THEN SHOWS **ONE** MAIN BUTTON. FOR A CONSOLE WITHOUT JELLYMATIC IT IS **INSTALL**.

4. PRESS IT, READ THE SMALL **APPROVE** WINDOW (IT NAMES THE EXACT IMAGE), PRESS **APPROVE**.
5. THE FIRST TIME, THE MANAGER TAKES THE FULL BACKUP OF BOTH CHIPS (ABOUT 15 MINUTES, NOTHING WRITTEN). THEN IT WRITES THE MENU, READS IT BACK, CHECKS THE WHOLE FLASH AND RESTARTS THE CONSOLE TO WATCH IT START. KEEP THE CABLE IN AND THE CONSOLE ON UNTIL IT SAYS **DONE**.
6. SWITCH THE CONSOLE OFF AND ON: THE START-UP MENU APPEARS. CARRY ON WITH FIRST RUN ON THE CONSOLE.





UPDATING LATER IS THE SAME: WHEN `release\launcher\` HOLDS A NEWER MENU, HOME SHOWS **UPDATE TO X.Y.Z**.

THE WINDOW

TAB	WHAT IS ON IT
HOME	THE CONSOLE'S STATE IN PLAIN WORDS AND AT MOST ONE MAIN BUTTON; UNDER **MORE**: CHECK AGAIN, OFFICIAL UPDATE, WI-FI, NEW JELLYFIN SIGN-IN
FIX A PROBLEM	SET UP JELLYFIN SOUND, BACK UP, REPAIR MENU, REMOVE MENU, SWITCH MENU OFF, RESTORE ORIGINAL, FORGET JELLYFIN, THE CONSOLE CHIP (FPGA) JOBS, FINISH JOB, CLOSE JOB (EACH BUTTON EXPLAINS ITSELF WHEN YOU HOVER)
REMOTE PILOT	SEE AND PRESS THE CONSOLE'S MENUS FROM THE PC (REMOTE PILOT)

TAB	WHAT IS ON IT
SETTINGS	AI CONTROL (OFF UNLESS YOU ALLOW IT), JELLYFIN ON THIS PC
ABOUT	FOLDERS, GUIDE, LICENCES

THE **EYE** BUTTON AT THE TOP HIDES PRIVATE DETAILS (ADDRESSES, NETWORK NAMES, USER NAMES, IDS) ON SCREEN, SO SCREENSHOTS ARE SAFE. IT IS ON AT EVERY START.

WI-FI AND JELLYFIN FROM THE PC (OPTIONAL)

YOU CAN SET UP EVERYTHING ON THE CONSOLE ITSELF (FIRST RUN). IF YOU PREFER THE PC: **MORE** › **WI-FI**. NETWORKS COME FROM `%LOCALAPPDATA%\Jellymatic\chromatic-wifi-networks.txt` (OPTIONAL; AN `ssid=` LINE THEN A `password=` LINE PER NETWORK, SEE `chromatic-wifi-networks.example.txt`) OR ARE TYPED IN THE WINDOW; THEY ARE **MERGED** WITH THE NETWORKS ALREADY ON THE CONSOLE (AT MOST 4). IF THIS PC HAS JELLYFIN SETTINGS (SETTINGS TAB), THE CONSOLE ALSO GETS **ITS OWN** JELLYFIN SIGN-IN THROUGH QUICK CONNECT: YOUR PASSWORD NEVER GOES TO THE CONSOLE, AND THE CONSOLE APPEARS AS ITS OWN DEVICE IN JELLYFIN'S DASHBOARD › DEVICES, WHERE YOU CAN REVOKE IT. JELLYMATIC ONLY EVER SIGNS OUT THE CONSOLE'S OWN SIGN-IN, NEVER YOUR OTHER DEVICES.

FULL BACKUP BEFORE ANY FPGA CHANGE

NOTHING IS WRITTEN TO A CONSOLE BEFORE A **FULL BACKUP OF BOTH CHIPS** EXISTS ON YOUR PC. THE FIRST TIME YOU PRESS **INSTALL** (OR **BACK UP**, AN OFFICIAL UPDATE, OR ANY CONSOLE CHIP JOB), THE MANAGER TAKES IT BY ITSELF, BEFORE ANYTHING ELSE, WITH NO EXTRA QUESTIONS:

1. IT CHECKS THE CONSOLE MODEL ON THE CHIPS THEMSELVES (THE ESP32, ITS FLASH CHIP, THE CONSOLE CHIP'S JTAG ID) AND REFUSES A MODEL OR FIRMWARE IT DOES NOT KNOW.
2. IT READS THE WHOLE **ESP32** FLASH (4 MIB) **TWICE**, EVERY 256 KIB PIECE CHECKED AGAINST THE CHIP'S OWN CHECKSUM, AND THE WHOLE **CONSOLE CHIP (FPGA)** FLASH (4 MIB) **TWICE** OVER ITS JTAG BRIDGE (NEEDS OPENFPGALOADER, SEE WHAT YOU NEED). THE TWO READS OF EACH CHIP

MUST BE IDENTICAL, OR NOTHING IS KEPT. BOTH CHIPS MUST BE CHECKED TO BELONG TO THE SAME CONSOLE.

3. THE CONSOLE MUST RUN THE **OFFICIAL MODRETRO FIRMWARE EXACTLY AS IT COMES**: NO START-UP MENU (NOT EVEN SWITCHED OFF), NOTHING HALF-INSTALLED, AND NONE OF JELLYMATIC'S CONSOLE CHIP IMAGES (SOUND 18.40, POCKET LINK 18.41). OTHERWISE THE BACKUP IS REFUSED, BECAUSE IT WOULD NOT BE YOUR ORIGINAL: PUT THE OFFICIAL FIRMWARE BACK WITH MODRETRO'S OWN UPDATER FIRST (RECOVERY).
4. IT SAVES THEM IN `%LOCALAPPDATA%\Jellymatic\chromatic-backup-<date>-<time>\`, MAKES THE THREE-FILE RESTORE BUNDLE, COPIES IT TO THE **SECOND COPY** FOLDER, AND READS EVERY FILE BACK.
5. ONLY THEN DOES IT RECORD THE BACKUP AS YOUR CONSOLE'S ORIGINAL AND GO ON.

THIS FIRST FULL BACKUP IS TAKEN **ONCE**. AN ORIGINAL THAT IS ALREADY RECORDED IS **NEVER REPLACED**: IF IT CANNOT BE CHECKED (FOR EXAMPLE THE SECOND COPY'S DISK OR USB STICK IS NOT CONNECTED), EVERY JOB THAT NEEDS IT STOPS AND SAYS SO, AND NOTHING IS READ OR WRITTEN. CONNECT THE DISK AND TRY AGAIN.

THIS TAKES ABOUT 15 MINUTES AND WRITES NOTHING TO THE CONSOLE (THE SCREEN GOES DARK FOR A MOMENT WHILE THE CONSOLE CHIP IS READ, THEN THE CONSOLE RESTARTS). YOUR WI-FI PASSWORDS AND JELLYFIN SIGN-IN ARE LEFT OUT OF EVERY BACKUP.

THE SECOND COPY IS USUALLY ON THE SAME DISK (`%USERPROFILE%\Jellymatic-Backups`). FOR REAL SAFETY PUT IT ON ANOTHER DISK OR A USB STICK: SET `JELLYMATIC_SECOND_COPY` TO A FOLDER THERE BEFORE THE FIRST BACKUP (FOR EXAMPLE `setx JELLYMATIC_SECOND_COPY "E:\Jellymatic-Backups"`, THEN OPEN A NEW WINDOW).

ADVANCED, THE SAME BACKUP BY HAND (READ-ONLY TOOLS, IN A POWERSHELL WINDOW IN THE JELLYMATIC FOLDER): `scripts\chromatic_backup.py mcu`, THEN `scripts\chromatic_backup.py fpga --folder "<the folder it made>"`, THEN `scripts\verify_chromatic_backup.py "<that folder>" --create-bundle --second-copy "<another disk>"`.

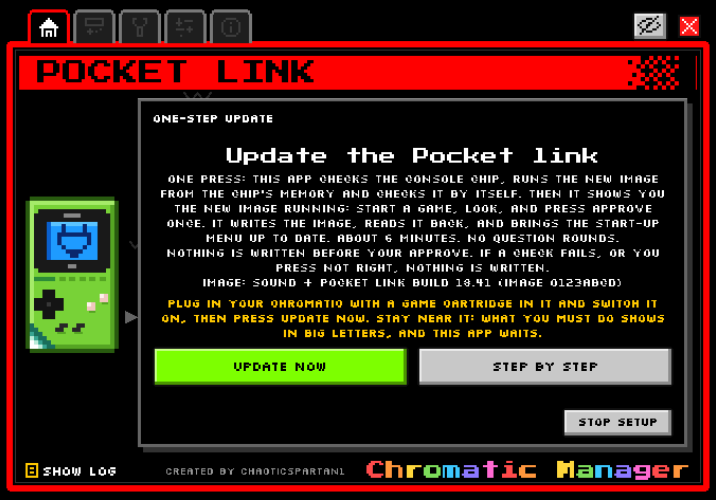
JELLYFIN SOUND (FPGA 18.40)

FIX A PROBLEM › SET UP JELLYFIN SOUND IS A GUIDED JOB: CHECK FPGA (READS ALL 4 MIB AND RECOGNISES YOUR ORIGINAL IMAGE), A REHEARSAL THAT WRITES YOUR ORIGINAL IMAGE BACK AND CHECKS IT (SO THE UNDO IS PROVEN ON YOUR CONSOLE FIRST), THEN THE SOUND IMAGE, READ BACK BYTE FOR BYTE. NEEDS THE FULL BACKUP ABOVE, OPENFPGALOADER, AND THE **OFFICIAL RELEASE FILES** `evt1_x2.fs` + `evt1_x2.bin` OF THE SOUND IMAGE IN `fpga\images\sound-18.40\build\` (PUBLISHED SEPARATELY AS RELEASE ASSETS, SEE WHAT YOU NEED).

UPDATE THE POCKET LINK IN ONE STEP (FPGA 18.41)

UPDATE THE POCKET LINK (HOME, WHEN THE CONSOLE HAS THE SOUND IMAGE AND THE REVIEWED 18.41 IMAGE IS IN `fpga\images\link-18.41\build\`) IS ONE SCREEN WITH ONE BUTTON, **UPDATE NOW**:

1. CHECK FPGA: THE FLASH MUST BE THE SOUND IMAGE, BYTE FOR BYTE.
2. THE NEW IMAGE IS LOADED INTO THE CHIP'S **RAM ONLY**. NOTHING IS WRITTEN YET.
3. AUTOMATIC CHECKS: THE IMAGE LOADED, USB CAME BACK, THE START-UP MENU REPORTS FPGA 18.41.
4. **YOUR APPROVE, WHILE THE NEW IMAGE RUNS**: START A GAME FOR A MOMENT AND CHECK PICTURE AND SOUND. *NOT RIGHT*, CANCEL OR SWITCHING THE CONSOLE OFF WRITES NOTHING (AN OFF-AND-ON RETURNS TO THE OLD IMAGE).
5. THE WRITE, WITH BACKUP CHECKS, FULL READ-BACK AND JOURNAL; THEN A RESTART FROM FLASH THAT MUST REPORT 18.41.
6. IF THE START-UP MENU IS OLDER THAN THE ONE IN `release\launcher\`, IT IS UPDATED TOO (ITS OWN APPROVE). THEN **READY: PLUG IN THE CABLE**, WITH **PUT THE PREVIOUS IMAGE BACK** ONE CLICK AWAY.



THE MANAGER ACCEPTS ONLY THE REVIEWED 18.41 IMAGE: ITS `.fs`, `.bin` AND STAGE RECEIPT ARE PINNED BY SHA-256 IN `scripts/chromatic_fpga.py`. ANY OTHER BUILD IS REFUSED, INCLUDING YOUR OWN BUILD OF THE SAME SOURCES (IT IS ALMOST NEVER BYTE-IDENTICAL), SO USE THE OFFICIAL RELEASE FILES.

TEXT MENU

`Chromatic-Manager.cmd` OPENS THE SAME TOOL AS A TEXT MENU: 1 STATUS, 2 BACKUP, 3 INSTALL, 4 REMOVE, 5 RESTORE-STOCK, 6 GUIDED OFFICIAL UPDATE, 7 WI-FI SETUP, 8 JELLYFIN RE-LINK, 9 JELLYFIN FORGET. EVERY WRITE ASKS YOU TO TYPE A ONE-TIME WORD SHOWN ON SCREEN (FOR EXAMPLE `yes482`).

WHERE YOUR FILES ARE

FOLDER	WHAT
`%LOCALAPPDATA%\Jellymatic`	EVERYTHING PERSONAL: JOURNALS, BACKUPS, LOGS, SETTINGS, WI-FI LIST, JELLYFIN SETTINGS
`%USERPROFILE%\Jellymatic-Backups`	THE SECOND COPY OF EVERY BACKUP. IT IS USUALLY ON THE SAME DISK AS THE FIRST: SET `JELLYMATIC_SECOND_COPY` TO ANOTHER DISK OR A USB STICK FOR REAL SAFETY

SET `JELLYMATIC_HOME` TO MOVE THE FIRST ONE. NONE OF IT IS EVER PART OF THE REPOSITORY. RUN `python scripts\user_paths.py` TO PRINT THE PATHS.

NEXT: FIRST RUN ON THE CONSOLE

FIRST RUN ON THE CONSOLE

JELLYMATIC / DOCS/FIRST-RUN-ON-THE-CONSOLE.MD

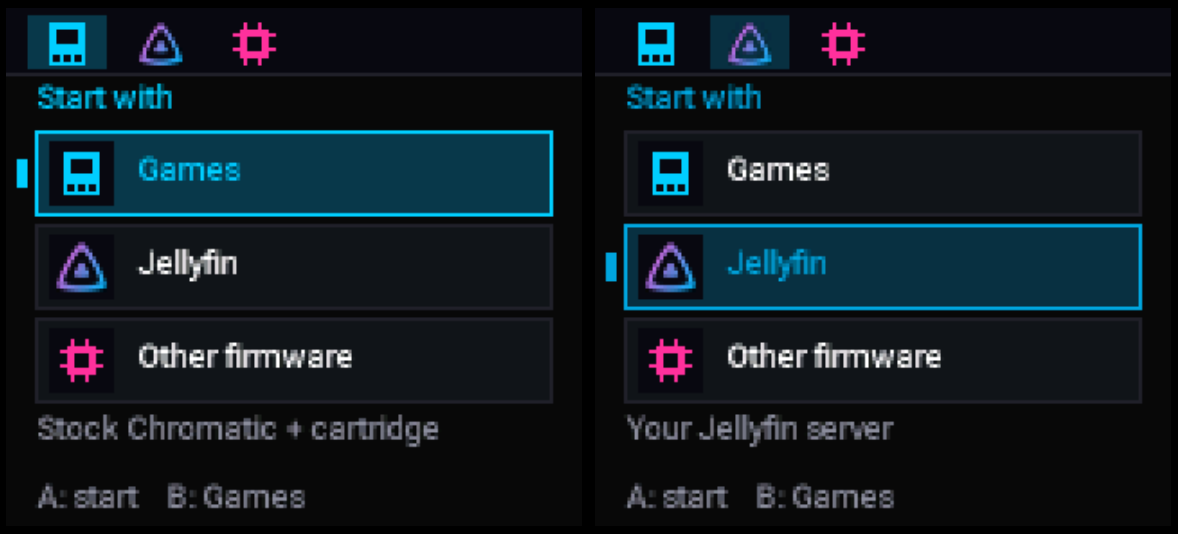
First run on the console

HOME › FIRST RUN ON THE CONSOLE

CONTENTS: OPEN JELLYFIN · ADD YOUR WI-FI · SIGN IN TO JELLYFIN · POCKET LINK · BUTTONS

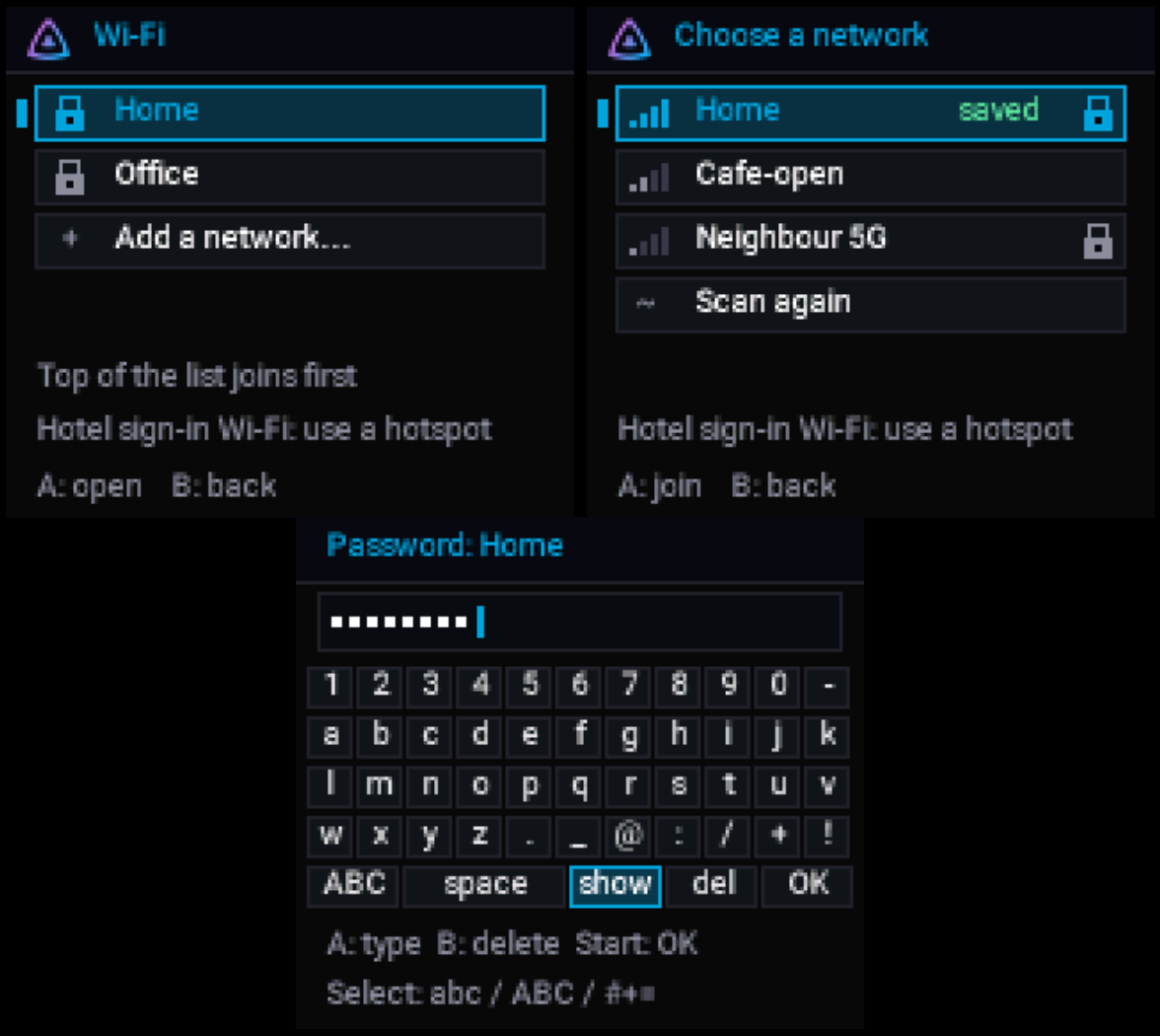
OPEN JELLYFIN

1. PUT A CARTRIDGE IN AND SWITCH THE CONSOLE **ON**. THE START-UP MENU APPEARS.
2. HIGHLIGHT **JELLYFIN** WITH THE D-PAD AND PRESS **A**.
3. PRESS **SELECT** TO OPEN **SETUP** (WI-FI NETWORKS / JELLYFIN SERVER).



ADD YOUR WI-FI

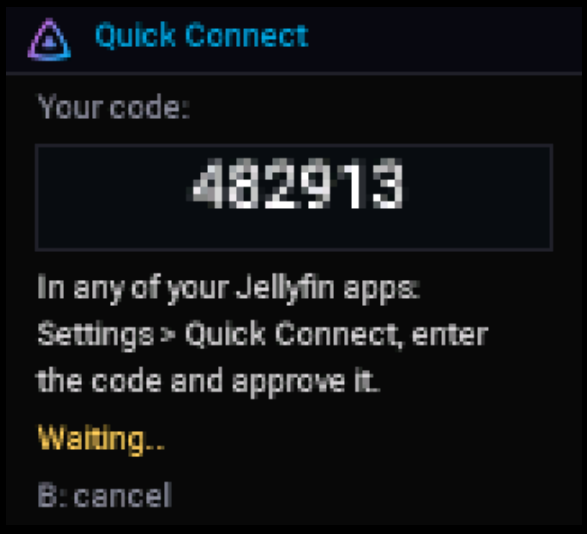
1. **SETUP › WI-FI NETWORKS › ADD A NETWORK…** THE CONSOLE SCANS AND LISTS NETWORKS NEARBY (STRONGEST FIRST; A LOCK MEANS A PASSWORD IS NEEDED).
2. CHOOSE YOURS AND TYPE THE PASSWORD ON THE ON-SCREEN KEYBOARD: D-PAD TO MOVE, **A** TYPES, **B** DELETES, **SELECT** SWITCHES ABC / abc / #+=, **START** IS OK.
3. THE CONSOLE SAYS **SAVED**, THEN **JOINING…**, THEN **CONNECTED**, OR THE REASON IT COULD NOT JOIN (WRONG PASSWORD, NETWORK NOT FOUND…) IN PLAIN WORDS.



UP TO **4** NETWORKS ARE KEPT. CHOOSE A SAVED ONE TO **USE IT NOW**, **CHANGE ITS PASSWORD** OR **FORGET** IT. 2.4 GHZ ONLY; HOTEL-STYLE SIGN-IN PAGES ARE NOT SUPPORTED (A PHONE HOTSPOT WORKS).

SIGN IN TO JELLYFIN (QUICK CONNECT)

1. **SETUP › JELLYFIN SERVER › FIND SERVERS NEARBY** (OR **ENTER SERVER ADDRESS**, FOR EXAMPLE `http://your-server:8096`).
2. CHOOSE YOUR SERVER. THE CONSOLE SHOWS A **6-DIGIT CODE**.
3. IN ANY JELLYFIN APP WHERE YOU ARE SIGNED IN (WEB, PHONE, TV): **SETTINGS › QUICK CONNECT**, ENTER THE CODE, APPROVE IT.
4. THE CONSOLE SAYS **SIGNED IN**. IT NOW HAS ITS OWN SIGN-IN, SHOWN IN JELLYFIN'S **DASHBOARD › DEVICES** AS "CHROMATIC". YOUR PASSWORD NEVER GOES TO THE CONSOLE.



QUICK CONNECT MUST BE ON IN THE SERVER'S **DASHBOARD › GENERAL**. TO UNDO: **SETUP › JELLYFIN SERVER › FORGET THIS SIGN-IN** (IT SIGNS THE CONSOLE OUT ON THE SERVER TOO), OR REVOKE THE DEVICE IN JELLYFIN.

POCKET LINK

ONLY WITH THE 18.41 CONSOLE CHIP IMAGE INSTALLED (INSTALL) AND POCKETFIN ON THE POCKET. SEE POCKET LINK.

BUTTONS

WHERE	BUTTONS
START-UP MENU	D-PAD CHOOSE · **A** START · **B** GAMES
JELLYFIN PAGE	**A** OPEN THE LIBRARY · **SELECT** SETUP · **B** OR **START** BACK TO THE MENU
SETUP SCREENS	D-PAD · **A** OPEN · **B** BACK
HOLDING ANY BUTTON WHILE SWITCHING ON	SKIPS THE MENU: STRAIGHT TO GAMES

NEXT: USING JELLYFIN ON THE CHROMATIC

USING JELLYFIN

JELLYMATIC / DOCS/USING-JELLYFIN-ON-THE-CHROMATIC.MD

Using Jellyfin on the Chromatic

HOME › USING JELLYFIN ON THE CHROMATIC

CONTENTS: BROWSING · PLAYING · SOUND · SLOW WI-FI · LEAVING · LIMITS

BROWSING

OPEN **JELLYFIN** FROM THE START-UP MENU. THE CONSOLE JOINS YOUR SAVED WI-FI BY ITSELF AND TALKS STRAIGHT TO YOUR JELLYFIN SERVER; NO PC IS NEEDED.

- **HOME**: ROWS IN JELLYFIN'S OWN ORDER (MY MEDIA, CONTINUE WATCHING, NEXT UP, LATEST PER LIBRARY).
- **FOLDERS**: GRIDS OF POSTERS OR EPISODE PICTURES.
- **DETAILS**: **RESUME** OR **FROM START**, HOW FAR YOU GOT, AND THE OVERVIEW.

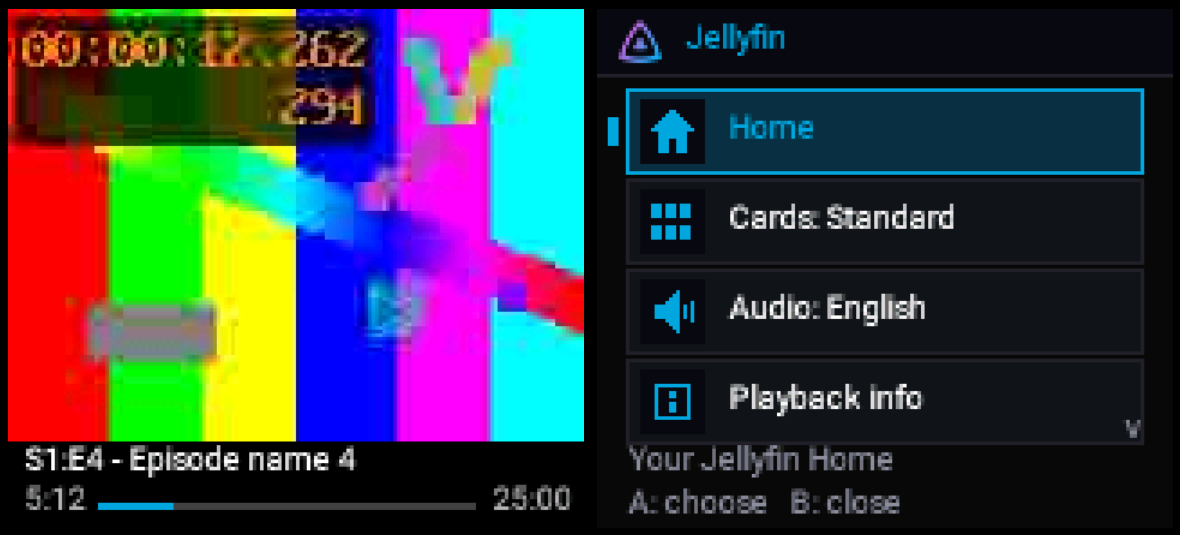




PLAYING

BUTTON	DOES
A	PAUSE / PLAY
LEFT / RIGHT	BACK / FORWARD 10 SECONDS
SELECT	SHOW OR HIDE THE BAR UNDER THE PICTURE

BUTTON	DOES
B	STOP



THE APP'S MENU HAS **HOME**, THE CARD SIZE, THE **AUDIO TRACK** (WHEN A TITLE HAS MORE THAN ONE) AND **PLAYBACK INFO**.

THE SERVER CONVERTS EACH TITLE TO A SIZE THE CONSOLE CAN SHOW (UP TO 160 × 120), AT THE TITLE'S OWN FRAME RATE. PICTURES THAT ARE LATE ARE SKIPPED RATHER THAN SHOWN OUT OF TIME. WHERE YOU STOPPED AND WHAT YOU WATCHED GO BACK TO THE SERVER, SO OTHER JELLYFIN APPS CONTINUE FROM THE SAME PLACE.

SOUND

SOUND NEEDS THE CONSOLE CHIP (FPGA) SOUND IMAGE 18.40 OR THE POCKET LINK IMAGE 18.41 (INSTALL). WITH THE OFFICIAL FPGA IMAGE, VIDEO PLAYS WITHOUT SOUND AND THE CONSOLE SAYS SO.

SLOW WI-FI

THE PLAYER STARTS AT ABOUT 1 MBIT/S AND UP TO 15 PICTURES A SECOND. IF THE NETWORK FALLS BEHIND IT HALVES THE BITRATE (DOWN TO 200 KBIT/S); IF THE CONSOLE FALLS BEHIND IT ASKS FOR FEWER PICTURES A SECOND. THE MESSAGE "ADJUSTING QUALITY" SHOWS WHEN IT DOES. SMOOTH PLAY EARNS THE QUALITY BACK NEXT TIME.

LEAVING

B STOPS PLAYBACK; **B** AGAIN GOES BACK THROUGH THE SCREENS; **START** ON THE JELLYFIN PAGE RETURNS TO THE START-UP MENU, WHERE **GAMES** STARTS YOUR CARTRIDGE.

LIMITS

- A CARTRIDGE MUST BE IN THE CONSOLE (THE APP DRAWS OVER IT).
- 2.4 GHZ WI-FI ONLY; NO HOTEL SIGN-IN PAGES.
- SUBTITLES AND SEARCH ARE NOT AVAILABLE YET.

NEXT: HOSTING JELLYFIN · POCKET LINK

HOSTING JELLYFIN



Hosting Jellyfin for wireless use

CONTENTS: WHAT THE CONSOLE NEEDS · STEPS · A SEPARATE SIGN-IN PER DEVICE · CHECKS · SAFETY

WHAT THE CONSOLE NEEDS

- A JELLYFIN SERVER ON THE **SAME LOCAL NETWORK** AS THE CONSOLE'S WI-FI.
- AN ADDRESS THE CONSOLE CAN REACH, FOR EXAMPLE `http://your-server:8096` (8096 IS JELLYFIN'S USUAL PORT).
- **QUICK CONNECT** TURNED ON, SO THE CONSOLE CAN GET ITS OWN SIGN-IN WITHOUT YOUR PASSWORD.
- THE SERVER'S FFMPEG (JELLYFIN'S OWN) ABLE TO TRANSCODE: THE CONSOLE ASKS FOR SMALL MJPEG VIDEO WITH ADPCM SOUND, WHICH THE SERVER MAKES ON THE FLY.

STEPS

1. INSTALL JELLYFIN ON A PC, NAS OR SMALL COMPUTER THAT STAYS ON: SEE JELLYFIN.ORG/DOWNLOADS. ADD YOUR LIBRARIES.
2. GIVE THE SERVER A **FIXED ADDRESS** ON YOUR NETWORK (IN YOUR ROUTER'S DHCP SETTINGS, "ADDRESS RESERVATION"), SO `http://your-server:8096` KEEPS WORKING.
3. ALLOW JELLYFIN THROUGH THE SERVER'S FIREWALL FOR YOUR **PRIVATE** NETWORK (PORT 8096, TCP), AND UDP 7359 IF YOU WANT THE CONSOLE'S **FIND SERVERS NEARBY** TO WORK.
4. IN JELLYFIN: **DASHBOARD › GENERAL › QUICK CONNECT: ENABLE**. SAVE.
5. CHECK THAT TRANSCODING WORKS: **DASHBOARD › PLAYBACK › TRANSCODING** (THE DEFAULT SETTINGS ARE FINE).
6. ON THE CONSOLE: **JELLYFIN › SELECT › JELLYFIN SERVER › FIND SERVERS NEARBY**, OR ENTER THE ADDRESS.

A SEPARATE SIGN-IN PER DEVICE (RECOMMENDED)

USE **QUICK CONNECT** FOR THE CONSOLE (AND FOR CHROMATIC MANAGER, WHEN YOU GIVE IT JELLYFIN SETTINGS). EACH GETS ITS **OWN** SIGN-IN, LISTED IN **DASHBOARD › DEVICES**, THAT YOU CAN REVOKE ON ITS OWN WITHOUT CHANGING YOUR PASSWORD OR SIGNING OUT YOUR TV OR PHONE. JELLYMATIC NEVER SIGNS OUT OTHER DEVICES.

YOU CAN ALSO MAKE A SEPARATE JELLYFIN **USER** FOR THE CONSOLE WITH ACCESS TO ONLY SOME LIBRARIES (DASHBOARD › USERS › ADD USER, THEN SIGN IN ON THE CONSOLE AS THAT USER).

CHECKS

PROBLEM	TRY
"FIND SERVERS NEARBY" FINDS NOTHING	ENTER THE ADDRESS BY HAND; ALLOW UDP 7359; CONSOLE AND SERVER ON THE SAME NETWORK
QUICK CONNECT CODE IS REFUSED	TURN QUICK CONNECT ON (DASHBOARD › GENERAL); THE CODE EXPIRES AFTER A FEW MINUTES
PLAYS, BUT STUTTERS	MOVE CLOSER TO THE ROUTER; THE CONSOLE ADJUSTS QUALITY BY ITSELF
NO SOUND	INSTALL THE SOUND IMAGE (INSTALL)

SAFETY

- KEEP JELLYFIN ON YOUR **HOME NETWORK**. JELLYMATIC DOES NOT NEED IT TO BE REACHABLE FROM THE INTERNET.
- IF YOU DO EXPOSE IT, USE HTTPS AND STRONG PASSWORDS; THE CONSOLE SUPPORTS `https://` ADDRESSES.
- NEVER POST YOUR SERVER ADDRESS, USER NAMES OR TOKENS IN ISSUES OR SCREENSHOTS.

NEXT: POCKET LINK

POCKET LINK



Pocket link

CONTENTS: WHAT IT IS · WHAT YOU NEED · SET IT UP · WHAT THE SCREEN SAYS · UNDO · LIMITS

WHAT IT IS

THE CHROMATIC JOINS YOUR WI-FI AND JELLYFIN; AN **ANALOGUE POCKET** PLUGGED INTO IT WITH A **LINK CABLE** THEN BROWSES YOUR LIBRARY AND PLAYS FROM IT, THROUGH THE CHROMATIC. THE POCKET SIDE IS **POCKETFIN** (ITS OWN REPOSITORY AND INSTRUCTIONS).

```
Jellyfin server ──Wi-Fi──► Chromatic (start-up menu, FPGA 18.41) ──link cable──► Analogue Pocket (PocketFin)
```

WHAT YOU NEED

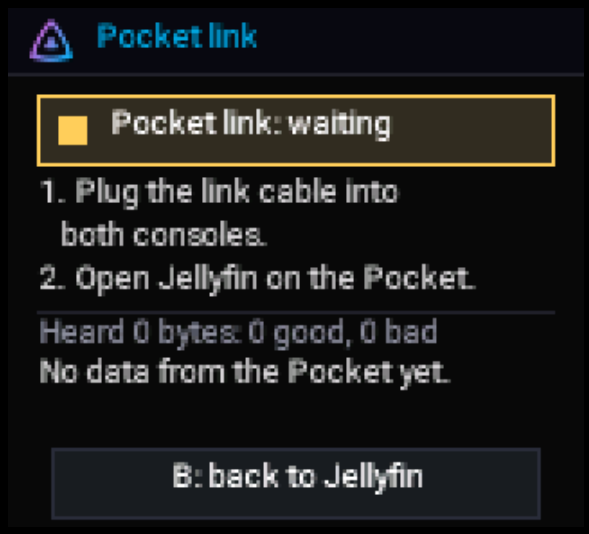
- THE START-UP MENU INSTALLED AND JELLYFIN SIGNED IN ON THE CHROMATIC (FIRST RUN).
- THE CONSOLE CHIP IMAGE **18.41** (SOUND + POCKET LINK) INSTALLED WITH UPDATE THE POCKET LINK. 18.41 IS MODRETRO'S OPEN-SOURCE FPGA DESIGN (V18.14) WITH SOUND HOOKS AND A LINK BRIDGE ADDED; EVERY GAME STILL PLAYS AS BEFORE.
- AN ANALOGUE POCKET WITH POCKETFIN INSTALLED, AND A LINK CABLE.

SET IT UP

1. ON THE CHROMATIC: START-UP MENU › **JELLYFIN**, AND STAY ON THE JELLYFIN PAGE. WITH 18.41 THE POCKET LINK STARTS THERE BY ITSELF AND SHOWS **POCKET LINK: WAITING**.
2. PLUG THE LINK CABLE INTO BOTH CONSOLES.
3. ON THE POCKET: OPEN POCKETFIN'S JELLYFIN CORE.

4. THE CHROMATIC SHOWS **POCKET LINK: CONNECTED**. BROWSE AND PLAY ON THE POCKET; KEEP THE CHROMATIC ON THAT SCREEN.





WHAT THE SCREEN SAYS

LIGHT	WORDS	MEANS
YELLOW	POCKET LINK: WAITING	PLUG THE CABLE IN, OPEN JELLYFIN ON THE POCKET
GREEN	POCKET LINK: CONNECTED	THE POCKET CAN BROWSE YOUR LIBRARY AND PLAY FROM HERE
YELLOW	SYNCING N OF M	COPYING PICTURES TO THE POCKET FOR OFFLINE BROWSING
GREEN	POCKET LINK: STREAMING	A VIDEO PLAYS ON THE POCKET
RED	POCKET LINK: UNPLUGGED	NO WORD FROM THE POCKET FOR 9 SECONDS: PLUG THE CABLE BACK IN
RED	POCKET LINK: STOPPED	THE LINK STOPPED (THE REASON IS SHOWN); IT TRIES AGAIN BY ITSELF
YELLOW	POCKET LINK: SLOW	THE FAST CONNECTION FAILED: BROWSE AND PICTURES ONLY, NO PLAYING ON THIS LINK

B GOES BACK TO THE JELLYFIN PAGE. **GAMES** ON THE START-UP MENU WORKS AS ALWAYS; THE LINK IS ONLY ACTIVE ON THE JELLYFIN PAGE.

UNDO

CHROMATIC MANAGER > **PUT THE PREVIOUS IMAGE BACK** (THE SOUND IMAGE 18.40), OR **FIX A PROBLEM > RESTORE FPGA** (YOUR OWN ORIGINAL FPGA IMAGE, FROM YOUR FULL BACKUP), OR MODRETRO'S OFFICIAL UPDATE TOOL. SEE RECOVERY.

LIMITS

- TESTED ON THE PC WITH SIMULATORS AND ON ONE CONSOLE: YOUR LINK CABLE AND POCKET MAY BEHAVE DIFFERENTLY.
- THE LINK SPEED IS LIMITED BY THE CABLE: PICTURE SIZE AND FRAME RATE ON THE POCKET ARE SMALL.

NEXT: REMOTE PILOT

Remote Pilot

HOME › REMOTE PILOT

CONTENTS: WHAT IT IS · WHY IT IS BUNDLED · IN CHROMATIC MANAGER · IN YOUR BROWSER · PAIRING · FOR AI ASSISTANTS · SAFETY · FULL DOCUMENTATION

WHAT IT IS

REMOTE PILOT IS A SMALL PROGRAM ON YOUR PC THAT SHOWS THE CHROMATIC IN YOUR WEB BROWSER: YOU **SEE ITS LIVE SCREEN, PRESS ITS BUTTONS, READ ITS LOGS AND SAVE SCREENSHOTS**, OVER THE USB CABLE OR YOUR HOME WI-FI. IT WORKS WITH THE START-UP MENU AND THE JELLYFIN SCREENS. IT CANNOT CHANGE FIRMWARE; ONLY CHROMATIC MANAGER DOES THAT.

THE `remote-pilot/` FOLDER IS A COPY OF THE REMOTEPILOT REPOSITORY: ITS CHROMATIC WEB TOOL PART (THE RECEIVER, THE WEB VIEWER, THE CHROMATIC PACKAGE AND THE **MODRETRO CHROMATIC** AND **EXAMPLE** PRESETS).

WHY IT IS BUNDLED

- **EASIER SET-UP AND TROUBLESHOOTING**: READ THE CONSOLE'S MENUS AND MESSAGES ON YOUR PC SCREEN INSTEAD OF SQUINTING AT THE HANDHELD, AND CLICK ITS BUTTONS.
- **CLEAN BUG REPORTS**: SCREENSHOTS AND LOGS WITH ADDRESSES, NETWORK NAMES AND CODES MASKED.
- **AI ASSISTANTS, SAFELY**: CLAUDE, CODEX OR ANOTHER ASSISTANT CAN TEST OR DEMO THE CONSOLE THROUGH REMOTE PILOT'S COMMAND LINE; IT PRESSES BUTTONS AND READS THE SCREEN, NOTHING MORE.
- **REMOTE DEMOS**: SHOW THE CONSOLE'S SCREENS TO SOMEONE ELSE FROM YOUR PC.

IT IS **OPTIONAL**, WORKS **ONLY ON YOUR OWN PC AND PRIVATE NETWORK** BY DEFAULT, AND IS **NEVER NEEDED** TO WATCH JELLYFIN.

IN CHROMATIC MANAGER

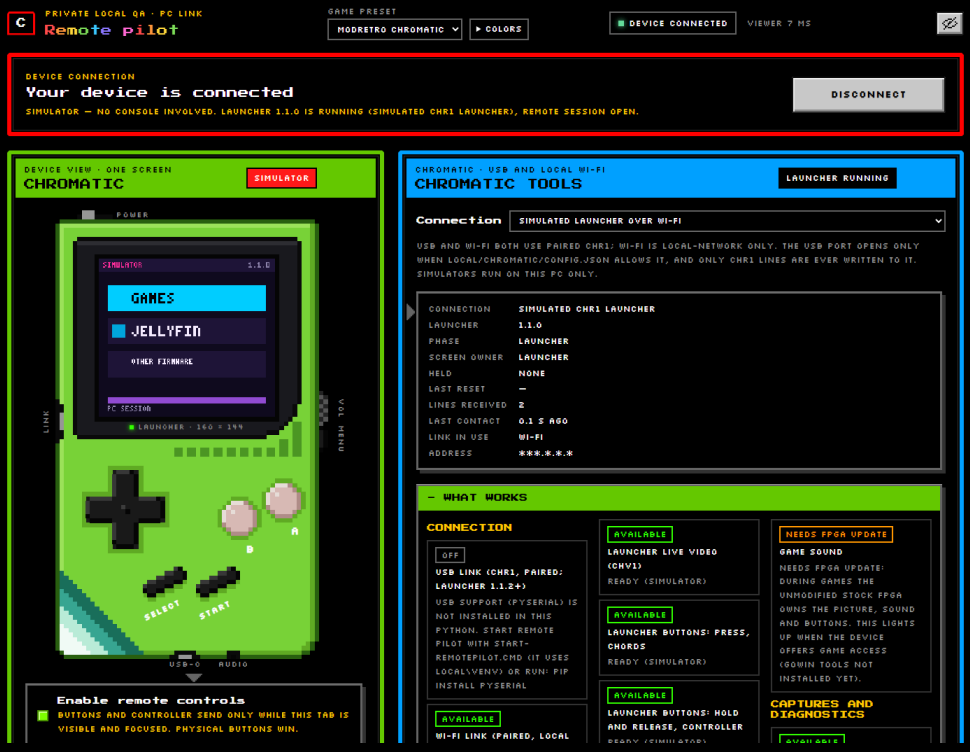
OPEN THE **REMOTE PILOT** TAB. THE MANAGER FINDS THE BUNDLED `remote-pilot\` FOLDER (OR THE FOLDER IN `REMOTE_PILOT_HOME`), STARTS ITS RECEIVER WITH REMOTE PILOT'S OWN PYTHON (`remote-pilot\local\venv`, MADE BY `Setup-Jellymatic.ps1`), AND OPENS THE CHROMATIC PAGE IN A WINDOW. WHILE A MANAGER JOB RUNS, REMOTE PILOT LETS GO OF THE USB PORT, AND GETS IT BACK AFTERWARDS.



IN YOUR BROWSER (WEB TOOL)

1. DOUBLE-CLICK `remote-pilot\Start-RemotePilot.cmd`.
2. OPEN `http://127.0.0.1:38764/` AND CHOOSE **MODRETRO CHROMATIC** IN THE **GAME PRESET** MENU.
3. PRESS **CONNECT TO DEVICE**. THE FIRST TIME, IT LISTS THE CONSOLE'S USB PORT AND OFFERS **ALLOW USB ON COMX**; ONLY THAT CLICK WRITES REMOTE PILOT'S PRIVATE SETTINGS (`remote-pilot\local\`, NEVER IN GIT).

TO TRY IT WITHOUT A CONSOLE, CHOOSE A **SIMULATED** CONNECTION IN THE CHROMATIC TOOLS PANEL.



PAIRING

- **PAIR OVER USB**: BOTH SCREENS SHOW THE SAME 6-DIGIT CODE; ONLY AN **A** PRESS ON THE CONSOLE APPROVES.
- OR LET CHROMATIC MANAGER'S WI-FI SETUP PROVISION A PAIRING KEY (IT WRITES THE CONSOLE AND `remote-pilot\local\chromatic\pairing.json` TOGETHER).
- WI-FI CONTROL WORKS ONLY ON YOUR PRIVATE NETWORK. A WI-FI PAIRING CODE EXPIRES AFTER 10 MINUTES, WORKS ONCE, AND IS REPLACED AFTER 20 WRONG TRIES. VIEWERS ON OTHER PCS OF YOUR NETWORK CAN WATCH, BUT CANNOT CHANGE THE MODE, PAIR, OR RUN MANAGER ACTIONS.

RUNNING CHROMATIC MANAGER FROM REMOTE PILOT (OPTIONAL, OFF BY DEFAULT)

REMOTE PILOT'S **FIRMWARE PATCHING** PANEL CAN START CHROMATIC MANAGER JOBS, BUT ONLY AFTER YOU SWITCH IT ON YOURSELF IN `remote-pilot\local\chromatic\config.json`:

```
{"managerProject": "C:\\path\\to\\Jellymatic", "managerRunAllowed": true,
 "managerPython": "C:\\path\\to\\LocalAppData\\Jellymatic\\venv\\Scripts\\python.exe"}
```

SET `managerPython` TO THE PYTHON THAT `Setup-Jellymatic.ps1` MADE (THE FULL PATH OF `%LOCALAPPDATA%\Jellymatic\venv\Scripts\python.exe`). EVERY WRITE STILL ASKS FOR

YOUR CONFIRMATION. EASIER AND RECOMMENDED: USE CHROMATIC MANAGER'S OWN WINDOW.

DETAILS: PAIRING WITH A CHROMATIC.

FOR AI ASSISTANTS (COMMAND LINE)

START REMOTE PILOT AND CONNECT FIRST, THEN (FROM THE `remote-pilot` FOLDER):

```
local\venv\Scripts\python.exe -m chromatic.cli status          # what is connected, launcher version
local\venv\Scripts\python.exe -m chromatic.cli press DOWN      # one short press (or a chord: press A B)
local\venv\Scripts\python.exe -m chromatic.cli shot --png      # a screenshot; prints its path
local\venv\Scripts\python.exe -m chromatic.cli tail --lines 20 # newest console log lines
```

EVERY COMMAND TALKS ONLY TO THE REMOTE PILOT ON THIS PC (127.0.0.1). MORE, WITH EXAMPLE PROMPTS: USING IT WITH AI ASSISTANTS AND AI PROMPT EXAMPLES.

SAFETY

- PASSWORD FIELDS ON THE CONSOLE NEVER SHOW THEIR TEXT WHILE REMOTE PILOT IS CONNECTED.
- REMOTE PILOT NEVER WRITES FLASH, NEVER PUTS THE CONSOLE INTO FIRMWARE-UPDATE MODE, AND NEVER TAKES THE PORT WHILE CHROMATIC MANAGER OR MODRETRO'S UPDATER WORKS.
- SCREENSHOTS CAN SHOW NETWORK NAMES: CHECK BEFORE YOU SHARE THEM.

FULL DOCUMENTATION

THE PUBLIC REPOSITORY **REMOTEPILOT** HAS THE FULL WIKI (ALSO BUNDLED IN `remote-pilot/docs/`): WHAT IT DOES, INSTALL, PAIRING, THE WEB VIEWER, AI ASSISTANTS, SECURITY MODEL, TROUBLESHOOTING AND FAQ.

NEXT: RECOVERY

RECOVERY

JELLYMATIC / DOCS/RECOVERY.MD

Recovery and troubleshooting

HOME › RECOVERY

CONTENTS: HOW JELLYMATIC AVOIDS BREAKING CHROMATIC OS · IF AN INSTALL STOPS HALF-WAY · WHAT THE RESULT MEANS · REINSTALL THE START-UP MENU · TURN THE MENU OFF OR REMOVE IT · PUT THE PREVIOUS FPGA IMAGE BACK · RESTORE THE ORIGINAL FPGA IMAGE · OFFICIAL UPDATES AND RESTORING CHROMATIC OS · COMMON PROBLEMS

HOW JELLYMATIC AVOIDS BREAKING CHROMATIC OS

- **THE OFFICIAL FIRMWARE STAYS.** THE START-UP MENU IS ADDED NEXT TO IT; THE OFFICIAL FIRMWARE SLOT, YOUR SETTINGS AND RADIO CALIBRATION ARE NEVER WRITTEN BY AN INSTALL.
- **GAMES IS ALWAYS ONE PRESS AWAY.** ANY RESTART THAT IS NOT A POWER-ON GOES STRAIGHT TO THE OFFICIAL FIRMWARE; HOLDING A BUTTON WHILE SWITCHING ON SKIPS THE MENU.
- **BACKUPS FIRST.** NOTHING IS WRITTEN BEFORE A VERIFIED FULL BACKUP OF BOTH CHIPS EXISTS (THE MANAGER TAKES IT BY ITSELF THE FIRST TIME), SAVED WITH A SECOND COPY. THE SECOND COPY IS USUALLY ON THE SAME DISK; PUT IT ON ANOTHER DISK OR A USB STICK FOR REAL SAFETY (INSTALL).
- **A JOURNAL.** EVERY STEP IS WRITTEN TO DISK BEFORE IT HAPPENS, SO AN INTERRUPTED JOB IS FINISHED OR UNDONE THE NEXT TIME THE MANAGER RUNS. NOTHING ELSE RUNS UNTIL THEN.
- **READ-BACK CHECKS.** EVERY WRITTEN REGION IS READ BACK FROM THE CHIP AND COMPARED, THEN THE WHOLE FLASH.
- **YOU APPROVE.** EVERY WRITE NEEDS YOUR **APPROVE** IN THE MANAGER'S WINDOW; FPGA UPDATES ARE FIRST RUN FROM THE CHIP'S RAM SO YOU CAN TRY A GAME **ON THE CONSOLE** BEFORE YOU APPROVE THE WRITE.
- **A START CHECK.** AFTER WRITING, THE CONSOLE IS RESTARTED AND WATCHED: IT MUST START AND KEEP RUNNING.
- **THE CHIP'S OWN RESCUE MODE.** THE ESP32'S DOWNLOAD MODE IS IN READ-ONLY ROM. NO FLASH CONTENT CAN BLOCK IT, SO THE MANAGER AND

MODRETRO'S OFFICIAL UPDATER CAN ALWAYS REACH THE CONSOLE.

IF AN INSTALL STOPS HALF-WAY

UNPLUGGED CABLE, FLAT BATTERY, CLOSED WINDOW: PLUG THE CONSOLE BACK IN, SWITCH IT **ON**, OPEN THE MANAGER. IT SEES THE JOURNAL AND OFFERS **FINISH JOB** (FINISH OR UNDO, FROM THE JOURNAL AND THE BACKUP). **CLOSE JOB** IS ONLY FOR A JOB THE MANAGER HAS CONFIRMED IS NO LONGER NEEDED. IF THE CONSOLE SHOWS NOTHING AT ALL, IT STILL WORKS: THE MANAGER REACHES IT THROUGH THE ROM DOWNLOAD MODE.

WHAT THE RESULT MEANS

RESULT	MEANING	WHAT TO DO
DONE (GREEN)	WRITTEN, READ BACK, CONSOLE SEEN STARTING	NOTHING
NOTHING WAS CHANGED (AMBER)	REFUSED BEFORE WRITING (FOR EXAMPLE A CONSOLE IT DOES NOT RECOGNISE)	READ THE REASON; NOTHING TO UNDO
PUT RIGHT	AN INTERRUPTED JOB WAS FINISHED OR UNDONE	NOTHING
STOPPED: THIS NEEDS YOU (RED)	IT STOPPED SAFELY AND TELLS YOU THE ONE THING TO DO	FOLLOW THE WORDS ON SCREEN, THEN FINISH JOB
…NOT SEEN RESTARTING	THE FLASH IS RIGHT, BUT THE CONSOLE WAS NOT SEEN STARTING	UNPLUG, SWITCH OFF AND ON, TRY A GAME

REINSTALL OR REPAIR THE START-UP MENU

FIX A PROBLEM › REPAIR MENU WRITES THE START-UP MENU AGAIN (SAME CHECKS AS INSTALL). USE IT WHEN THE MENU IS MISSING, DAMAGED OR SWITCHED OFF.

TURN THE MENU OFF OR REMOVE IT

BUTTON	WHAT IT DOES	YOUR WI-FI/JELLYFIN SET-UP
SWITCH MENU OFF	ERASES ONLY THE MENU PROGRAM (POWER-SAFE); THE CONSOLE BOOTS STRAIGHT TO THE OFFICIAL FIRMWARE	KEPT, FOR WHEN YOU INSTALL AGAIN
REMOVE MENU	PUTS THE OFFICIAL BOOT PROGRAM AND PARTITION TABLE BACK AND ERASES THE MENU AREA	ERASED; THE CONSOLE'S JELLYFIN SIGN-IN IS SIGNED OUT ON THE SERVER
RESTORE ORIGINAL	REWRITES THE ESP32 FROM YOUR NEWEST VERIFIED BACKUP OF THE OFFICIAL FIRMWARE (SETTINGS KEPT). THE FIRST OF THOSE BACKUPS IS YOUR ORIGINAL, TAKEN ONLY FROM THE OFFICIAL FIRMWARE AS IT COMES	ERASED; SIGN-IN SIGNED OUT

AFTER **REMOVE MENU** OR **RESTORE ORIGINAL** THE CONSOLE IS EXACTLY THE OFFICIAL LAYOUT. INSTALL BRINGS THE MENU BACK AT ANY TIME.

PUT THE PREVIOUS FPGA IMAGE BACK

AFTER **UPDATE THE POCKET LINK**, **PUT THE PREVIOUS IMAGE BACK** (ONE APPROVE) WRITES THE SOUND IMAGE 18.40 AGAIN, READ BACK BYTE FOR BYTE.

RESTORE THE ORIGINAL FPGA IMAGE

FIX A PROBLEM › RESTORE FPGA WRITES YOUR **OWN** ORIGINAL FPGA IMAGE FROM YOUR FULL BACKUP, READS ALL 4 MIB BACK AND COMPARES THEM, THEN RESTARTS THE CHIP. NOTHING FROM ANYONE ELSE'S CONSOLE IS EVER USED. THAT ORIGINAL IS TAKEN ONCE, BY THE FIRST FULL BACKUP, AND ONLY FROM MODRETRO'S OWN CONSOLE CHIP IMAGE (NEVER FROM JELLYMATIC'S SOUND OR POCKET LINK IMAGE); IT IS NEVER REPLACED AFTERWARDS.

IF THE MANAGER SAYS YOUR FULL BACKUP **COULD NOT BE CHECKED IN BOTH COPIES**, CONNECT THE DISK OR USB STICK THAT HOLDS THE SECOND COPY AND

TRY AGAIN. IT NEVER TAKES A NEW ORIGINAL OVER AN EXISTING ONE. IF THE FIRST BACKUP IS REFUSED BECAUSE THE CONSOLE IS **NOT ON THE OFFICIAL FIRMWARE AS IT COMES**, PUT THE OFFICIAL FIRMWARE BACK WITH MODRETRO'S UPDATER (BELOW), THEN PRESS **BACK UP**.

OFFICIAL UPDATES AND RESTORING CHROMATIC OS

USE **MODRETRO'S OFFICIAL UPDATE TOOL**, DOWNLOADED FROM MODRETRO (MODRETRO.COM). JELLYMATIC NEVER SHIPS IT. THE MANAGER GUIDES YOU (HOME › MORE › **OFFICIAL UPDATE**, OR TEXT MENU CHOICE 6):

1. THE MANAGER SWITCHES THE START-UP MENU OFF (POWER-SAFE).
2. YOU RUN MODRETRO'S UPDATER: **FIRMWARE UPDATE ONLY**, NOT CART CLINIC. KEEP THE CONSOLE PLUGGED IN AND ON. WHEN IT REPORTS SUCCESS, CLOSE IT AND PRESS **DONE** IN THE MANAGER.
3. THE MANAGER CHECKS WHAT CHANGED (THE UPDATER DOES NOT CHECK ITS OWN ESP32 WRITE), BACKS UP THE NEW FIRMWARE, AND SWITCHES THE START-UP MENU BACK ON (**INSTALL**).

GOOD TO KNOW:

- THE OFFICIAL UPDATER WRITES THE FPGA FIRST, THEN THE WHOLE ESP32 THROUGH THE ROM DOWNLOAD MODE, SO **IT ALWAYS WORKS**, WHATEVER JELLYMATIC LEFT ON THE CONSOLE. THAT IS YOUR LAST-RESORT RESTORE OF CHROMATIC OS: RUN IT, AND THE CONSOLE IS FULLY OFFICIAL AGAIN.
- IT **ERASES YOUR CONSOLE SETTINGS** (BRIGHTNESS, PALETTE AND SO ON); THAT IS THE OFFICIAL UPDATER'S BEHAVIOUR, NOT JELLYMATIC'S.
- IT ALSO REPLACES THE FPGA IMAGE WITH THE OFFICIAL ONE, SO JELLYFIN SOUND AND THE POCKET LINK ARE GONE UNTIL YOU INSTALL THEM AGAIN. THE START-UP MENU STAYS ON THE FLASH, SWITCHED OFF, UNTIL **INSTALL**.
- IF THE UPDATER FAILS: UNPLUG USB, SWITCH THE CONSOLE OFF AND ON, RUN IT AGAIN. IT NEEDS THE INTERNET AND MAY INSTALL THE GOWIN USB DRIVER ON WINDOWS.

TO REINSTALL JELLYMATIC ON TOP AFTERWARDS: OPEN THE MANAGER, PRESS **INSTALL**, THEN (OPTIONAL) **SET UP JELLYFIN SOUND** AND **UPDATE THE POCKET LINK** AGAIN.

COMMON PROBLEMS

PROBLEM	TRY
MANAGER SAYS "PLUG IN"	USE A DATA CABLE, SWITCH THE CONSOLE ON, CLOSE OTHER TOOLS THAT USE THE PORT (MODRETRO'S UPDATER, A SERIAL MONITOR)
"NOT THE CONSOLE THE BACKUPS CAME FROM"	CONNECT YOUR OWN CONSOLE, THE ONE THE BACKUPS WERE MADE FROM
"NOT A LAYOUT OR FIRMWARE THIS TOOL KNOWS"	YOUR CONSOLE HAS FIRMWARE JELLYMATIC WAS NOT TESTED WITH. NOTHING WAS WRITTEN. USE THE OFFICIAL UPDATER FIRST, OR WAIT FOR A NEWER JELLYMATIC
MENU DOES NOT APPEAR	IS A CARTRIDGE IN? ARE YOU HOLDING A BUTTON AT POWER-ON?
GAMES WILL NOT START FROM THE MENU	LET GO OF ALL BUTTONS; INSERT A CARTRIDGE

SEE ALSO THE FAQ.

FAQ

HOME › FAQ

CONTENTS: GENERAL · SAFETY · JELLYFIN · POCKET LINK · HELP

GENERAL

IS THIS MADE BY MODRETRO OR JELLYFIN? NO. JELLYMATIC IS AN INDEPENDENT PROJECT, NOT AFFILIATED WITH MODRETRO OR THE JELLYFIN PROJECT.

DOES IT COST ANYTHING? NO. IT IS FREE SOFTWARE (GPL-3.0-OR-LATER). BUILDING THE FPGA IMAGES NEEDS GOWIN'S FPGA TOOLS, WHICH NEED A (FREE) LICENCE FROM GOWIN.

WHICH CONSOLES? THE MODRETRO CHROMATIC. TESTED WITH OFFICIAL FIRMWARE MCU V4.6 AND FPGA V18.14 ON ONE CONSOLE. THE MANAGER REFUSES FIRMWARE OR LAYOUTS IT DOES NOT RECOGNISE.

DO I NEED THE PC AFTER SET-UP? NO. WI-FI AND JELLYFIN ARE SET UP ON THE CONSOLE AND IT TALKS TO YOUR SERVER DIRECTLY. THE PC IS FOR INSTALLING, UPDATING AND RECOVERY.

SAFETY

CAN IT BREAK MY CONSOLE? EVERY WRITE IS BACKED UP, JOURNALLED, READ BACK AND APPROVED BY YOU, AND THE OFFICIAL FIRMWARE STAYS ON THE CONSOLE. THE ESP32'S ROM RESCUE MODE CANNOT BE BLOCKED BY ANYTHING IN FLASH, SO THE MANAGER AND MODRETRO'S UPDATER CAN ALWAYS REACH IT. STILL: **USE IT AT YOUR OWN RISK.**

ARE MY GAMES AND SAVES SAFE? JELLYMATIC NEVER READS OR WRITES CARTRIDGES, FLASH CARTS, THEIR SD CARDS OR SAVES. **GAMES** STARTS THE UNTOUCHED OFFICIAL FIRMWARE.

CAN I GO BACK TO COMPLETELY OFFICIAL? YES: **REMOVE MENU** OR **RESTORE ORIGINAL** IN THE MANAGER, OR MODRETRO'S OFFICIAL UPDATE TOOL (RECOVERY).

DOES IT SEND ANYTHING ANYWHERE? NO. NOTHING IS UPLOADED. THE CONSOLE TALKS ONLY TO YOUR JELLYFIN SERVER (AND, IF YOU USE IT, REMOTE PILOT ON YOUR OWN PC).

JELLYFIN

WHERE IS MY JELLYFIN PASSWORD? ONLY WHERE YOU PUT IT. THE CONSOLE GETS ITS OWN SIGN-IN WITH QUICK CONNECT; THE MANAGER CAN DO THE SAME IF YOU GIVE IT THIS PC'S JELLYFIN SETTINGS (KEPT IN `%LOCALAPPDATA%\Jellymatic`).

WHY IS THERE NO SOUND? SOUND NEEDS THE FPGA SOUND IMAGE 18.40 OR 18.41 (INSTALL).

WHY IS THE PICTURE SMALL? THE CONSOLE'S SCREEN IS 160 × 144; VIDEO IS UP TO 160 × 120.

POCKET LINK

WHAT DO I NEED? AN ANALOGUE POCKET WITH POCKETFIN, A LINK CABLE, AND THE 18.41 FPGA IMAGE ON THE CHROMATIC (POCKET LINK).

DOES THE POCKET NEED WI-FI? NO. THE CHROMATIC DOES THE NETWORK PART.

HELP

WHERE DO I REPORT A BUG? IN THE REPOSITORY'S ISSUES (CONTRIBUTING). SECURITY PROBLEMS PRIVATELY (SECURITY). SUPPORT IS NOT GUARANTEED.

CAN I USE AN AI ASSISTANT? YES, WITH CARE: AI PROMPT EXAMPLES.

BUILDING FROM SOURCE



JELLYMATIC / DOCS/BUILDING-FROM-SOURCE.MD

Building from source

HOME › BUILDING FROM SOURCE

CONTENTS: START-UP MENU · CONSOLE CHIP (FPGA) IMAGES · CHROMATIC MANAGER · TESTS · PREVIEWS AND PICTURES · OFFLINE DOCUMENTATION PAGE

EVERYTHING BUILDS ON WINDOWS. NOTHING IN THIS PAGE TOUCHES A CONSOLE.

START-UP MENU

NEEDS: GIT, PYTHON 3.12, ABOUT 3 GB OF DISK.

```
powershell -ExecutionPolicy Bypass -File scripts\Get-Upstreams.ps1 -EspIdf   # ESP-IDF v5.3
at the pinned commit
$env:IDF_TOOLS_PATH = "$env:LOCALAPPDATA\Jellymatic\tools\esp32-tools"
upstream\esp-idf\install.ps1 esp32                                            # Espressif's
own tool installer
powershell -ExecutionPolicy Bypass -File scripts\Build-ChromaticLauncher.ps1
```

THE RESULT GOES TO `release\launcher\` (IGNORED BY GIT), WHERE CHROMATIC MANAGER INSTALLS IT FROM. `idf.py flash` IS MADE TO FAIL ON PURPOSE: IT WOULD OVERWRITE THE CONSOLE'S OFFICIAL FIRMWARE. INSTALL ONLY WITH CHROMATIC MANAGER.

SETTINGS: `chromatic\launcher\CMakeLists.txt` (`PROJECT_VER`, `PJ_ENABLE_SETUP`), `chromatic\launcher\sdkconfig.defaults`, `chromatic\launcher\partitions.csv` (THE THREE OFFICIAL ROWS FIRST, UNCHANGED).

CONSOLE CHIP (FPGA) IMAGES

NEEDS: GOWIN EDA **V1.9.12.04** WITH **YOUR OWN** GOWIN LICENCE (FROM GOWINSEMI.COM; NEVER COMMIT IT), AND MODRETRO'S FPGA SOURCE:

```
powershell -ExecutionPolicy Bypass -File scripts\Get-Upstreams.ps1 -ChromaticFpga
$env:GOWIN_SH = "C:\Gowin\Gowin_V1.9.12.04_x64\IDE\bin\gw_sh.exe"              # where your
Gowin is
py -3 scripts\stage_chromatic_fpga_link.py --build                            # 18.41: stage
+ Gowin build
py -3 scripts\stage_chromatic_fpga_link.py --export my-18.41                  # copy image,
receipts, sources
```

THE STAGING SCRIPTS COPY MODRETRO'S TREE (NEVER EDITED) TO `staging\` AND APPLY EXACT TEXT PATCHES; EVERY ANCHOR MUST MATCH OR THEY STOP. `scripts\stage_chromatic_fpga_media.py` DOES THE SAME FOR THE SOUND IMAGE 18.40 (BUILD IT WITH `gw_sh build.tcl` IN `staging\chromatic-fpga-media\esp32t`, THEN `py -3 scripts\record_chromatic_fpga_build.py staging\chromatic-fpga-media\esp32t`).

`fpga\images\link-18.41\` AND `fpga\images\sound-18.40\` HOLD THE **EXACT SOURCES AND RECEIPTS** OF THE REVIEWED IMAGES, BUT NOT THE IMAGES THEMSELVES. TO USE AN IMAGE, PUT THE **OFFICIAL RELEASE FILES** `evt1_x2.fs` AND `evt1_x2.bin` (PUBLISHED SEPARATELY AS RELEASE ASSETS) IN THAT FOLDER'S `build\`. THE MANAGER REFUSES ANY IMAGE WHOSE BYTES DIFFER FROM ITS RECEIPT (AND, FOR 18.41, FROM THE SHA-256 PINS IN `scripts\chromatic_fpga.py`). A BUILD OF YOUR OWN IS ALMOST NEVER BYTE-IDENTICAL (ANY OTHER GOWIN VERSION, SETTING OR IP CHANGES THE BYTES), SO IT WILL NOT MATCH THE PINS AND IS REFUSED; THAT IS ON PURPOSE, BECAUSE ONLY THE REVIEWED IMAGE WAS TRIED ON A CONSOLE. BUILDING IS STILL USEFUL TO CHECK THE SOURCES AND THE TIMING REPORTS.

THE RTL ADDED BY JELLYMATIC IS IN `rtl\` (POCKET LINK BRIDGE, LINK-PORT BYTE/STREAM, FIFOS) AND `chromatic\fpga-media\` (SOUND HOOKS), EACH WITH ICARUS VERILOG TEST BENCHES.

CHROMATIC MANAGER

RUNS FROM SOURCE (`Setup-Jellymatic.ps1`, THEN `Chromatic-Manager-Window.cmd`). TO MAKE A PORTABLE WINDOWS PROGRAM FOR YOURSELF: `scripts\Build-ChromaticManagerApp.ps1` (PYINSTALLER; OUTPUT IN `dist\`). IF YOU **SHARE** SUCH A BUILD, YOU TAKE ON THE OBLIGATIONS OF THE BUNDLED LIBRARIES: QT/PYSIDE6 IS LGPL-3.0 (KEEP THE QT LIBRARIES REPLACEABLE, SHIP THEIR LICENCE AND POINT TO THEIR SOURCE), ESPTOOL IS GPL-2.0-OR-LATER, PYTHON AND THE OTHER PACKAGES HAVE THEIR OWN LICENCES. THAT IS WHY THIS REPOSITORY SHIPS SOURCE, NOT A BUILT PROGRAM.

TESTS

NEEDS MSYS2 UCRT64 GCC (`C:\msys64\ucrt64\bin\gcc.exe`) FOR THE C HOST CHECKS, AND FOR THE RTL BENCHES ICARUS VERILOG IN `<tools>\fpga-tools\native\ucrt64` (`bin\iverilog.exe`, `bin\vvp.exe`; `<tools>` IS `%LOCALAPPDATA%\Jellymatic\tools` OR `JELLYMATIC_TOOLS`).

```
$py = "$env:LOCALAPPDATA\Jellymatic\venv\Scripts\python.exe"
& $py -B scripts\test_chromatic_manager.py        # Manager logic against a simulated
console
& $py -B scripts\test_chromatic_manager_gui.py    # the window, offscreen
& $py -B scripts\test_chromatic_launcher.py       # launcher host checks (+ ELF checks when
built)
& $py -B scripts\test_chromatic_setup.py          # on-console Wi-Fi/Jellyfin set-up,
power-cut model
& $py -B scripts\test_chromatic_fpga.py           # FPGA jobs against a simulated chip
```

BUILD THE START-UP MENU FIRST (ABOVE): THE MANAGER TESTS INSTALL THAT BUILD ON A SIMULATED CONSOLE, AND THE LAUNCHER TEST CHECKS THE LINKED PROGRAM; WITHOUT IT THEY STOP WITH "NO LAUNCHER RELEASE FOUND" (THE LAUNCHER TEST SKIPS ITS BUILD CHECKS). THE FPGA STAGING TESTS NEED MODRETRO'S SOURCE (`Get-Upstreams.ps1 -ChromaticFpga`) AND STAGED TREES: RUN `py -3 scripts\stage_chromatic_fpga_media.py`, `py -3 scripts\check_pocket_link_patch.py` AND `py -3 scripts\stage_chromatic_fpga_link.py` FIRST (NOTHING IS BUILT BY THESE; THEY ONLY COPY AND PATCH).

THE TESTS NEVER OPEN A DEVICE. WHERE THEY NEED A STOCK FIRMWARE IMAGE THEY USE A **MADE-UP** ONE (`compose_chromatic_image.synthetic_stock`, NO MODRETRO CODE IN IT), UNLESS YOU POINT `JELLYMATIC_STOCK_DUMP` AT A FULL READ OF YOUR OWN CONSOLE. REMOTE PILOT'S TESTS ARE IN `remote-pilot\` (`python -m unittest test_chromatic`, `node test_chromatic_panel.js`).

PREVIEWS AND PICTURES

- `scripts\render_chromatic_manager_previews.py --out <folder>`: THE MANAGER WINDOW IN EACH STATE, WITH MADE-UP DATA.
- `scripts\test_chromatic_launcher.py` AND `test_chromatic_setup.py` WRITE THE CONSOLE'S SCREENS AS `.ppm` / `.png` PREVIEWS UNDER `staging\`.
- `scripts\test_chromatic_player.py` WRITES THE JELLYFIN APP'S SCREENS (SYNTHETIC TEST PATTERN VIDEO).

- THE CONSOLE'S TEXT FONT, `pocket\ui_font.h`, IS MADE FROM GOOGLE'S ROBOTO MEDIUM (APACHE-2.0) BY `py -3 scripts\build_ui_font.py --fetch` (NEEDS `fonttools`; THE FONT IS PINNED BY SHA-256 IN `upstream.lock.json`). `--check` ONLY COMPARES THE RESULT WITH THE FILE IN THE REPOSITORY.
- THE README'S BANNER, SECTION HEADERS AND DIVIDERS ARE PICTURES IN `docs/images/readme/` (GITHUB DOES NOT SHOW PAGE STYLES, SO THE PIXEL LOOK TRAVELS AS IMAGES); KEEP A NORMAL `##` HEADING UNDER EACH HEADER PICTURE FOR SCREEN READERS AND LINKS.

OFFLINE DOCUMENTATION PAGE

`Open-Documentation.html` IN THE MAIN FOLDER IS EVERY DOCUMENTATION PAGE IN ONE FILE THAT OPENS BY DOUBLE-CLICK, OFFLINE. **AFTER EDITING ANY `.md` PAGE, REBUILD IT:**

```
py -3 scripts\build_docs_html.py          # rewrites Open-Documentation.html
py -3 scripts\test_docs_html.py           # fails if it is out of date or would load
anything from the internet
```

IT NEEDS ONLY PYTHON (NO PACKAGES). IT CONVERTS README, `docs\*.md`, CONTRIBUTING, SECURITY, THIRD-PARTY-NOTICES AND ADDITIONAL-TERMS (ONLY FILES TRACKED BY GIT) AND PUTS THEM INSIDE THE PAGE; THE PICTURES AND THE PIXEL FONTS (`docs\fonts\`, SIL OFL 1.1) LOAD FROM THE REPOSITORY FOLDER. A NEW PAGE IN `docs\` APPEARS UNDER **MORE** UNTIL YOU ADD IT TO A TAB IN `GROUPS` AT THE TOP OF THE SCRIPT. THE SAME RUN WRITES `docs\index.html`, A COPY FOR GITHUB PAGES (SEE BELOW).

THE PDF GUIDE (`docs\Jellymatic-Guide.pdf`, SHOWN BY GITHUB IN THE BROWSER) IS PRINTED FROM THE SAME PAGES BY CHROME OR EDGE, OFFLINE:

```
py -3 scripts\build_docs_pdf.py           # rewrites docs\Jellymatic-Guide.pdf and
docs\.guide-source.sha256
```

THE TEST ABOVE SAYS WHEN THE PDF IS OUT OF DATE (IT COMPARES `docs\.guide-source.sha256` WITH THE PAGES AND BUILD SCRIPTS; IT DOES NOT PRINT AGAIN). SET `DOCS_BROWSER` IF CHROME OR EDGE IS NOT FOUND.

FOR MAINTAINERS, WHEN THE REPOSITORY IS PUBLIC: GITHUB CAN HOST THE DOCUMENTATION FOR FREE. IN THE REPOSITORY'S SETTINGS > PAGES, CHOOSE DEPLOY FROM A BRANCH, BRANCH `main`, FOLDER `/docs`. THE SITE IS THEN

`https://chaoticspartan1.github.io/Jellymatic/` (IT SERVES `docs\index.html`; `docs\.nojekyll` MAKES GITHUB SERVE THE FILES AS THEY ARE). ADD A LINK TO IT IN THE README ONLY ONCE IT WORKS.

AI PROMPT EXAMPLES

AI prompt examples (Claude, Codex and other assistants)

HOME › AI PROMPT EXAMPLES

CONTENTS: SAFETY RULES · CONNECT AN ASSISTANT · PROMPTS THAT ONLY READ · PROMPTS THAT MAY CHANGE THE CONSOLE · PROMPTS FOR DEVELOPERS · BUG REPORTS

SAFETY RULES (READ FIRST)

1. **YOU APPROVE EVERY CHANGE.** THE MANAGER'S API LETS AN ASSISTANT *READ* FREELY AND *ASK* FOR A JOB; EVERY WRITE WAITS FOR **YOUR APPROVE** IN THE MANAGER'S WINDOW. AI CONTROL IS **OFF** UNLESS YOU TURN IT ON (SETTINGS › AI CONTROL), AND **STOP AI NOW** ENDS IT AT ONCE.
2. **NEVER GIVE AN ASSISTANT YOUR SECRETS.** DO NOT PASTE WI-FI PASSWORDS, JELLYFIN PASSWORDS, TOKENS, SERVER ADDRESSES OR BACKUPS INTO A CHAT. THE MANAGER NEVER RETURNS THEM THROUGH ITS API.
3. **BE PRESENT.** ONLY LET AN ASSISTANT START A CONSOLE JOB WHEN YOU ARE AT THE CONSOLE, IT IS PLUGGED IN AND ON, AND YOU CAN SWITCH IT OFF AND ON IF ASKED.
4. **NO FLASHING BY HAND.** ASK ASSISTANTS TO USE CHROMATIC MANAGER ONLY, NEVER `esptool write_flash`, `idf.py flash` OR `openFPGALoader --write-flash` DIRECTLY. (THE PROJECT MAKES `idf.py flash` FAIL ON PURPOSE.)
5. **ENOUGH BUDGET LEFT.** AI-STARTED WRITES STAY **REFUSED** UNLESS YOU TURN AI CONTROL ON (RULE 1), AND EVEN THEN EACH ONE ALSO NEEDS THE FULL BACKUP AND A FRESH *USAGE READING*, SO AN ASSISTANT THAT IS ABOUT TO RUN OUT OF ITS LIMITS DOES NOT START A FIRMWARE CHANGE IT CANNOT FINISH. THE ASSISTANT (OR YOU) WRITES THE READING TO `%LOCALAPPDATA%\Jellymatic\chromatic-tools\usage-snapshot.json` JUST BEFORE IT ASKS:

```
{"observed_at": "2026-11-10T14:02:00+00:00", "used_percent_by_window": [41, 12],
 "reporter": "claude", "ordinary_usage_allowed": true}
```

- `observed_at`: WHEN THE USAGE WAS READ, WITH ITS TIME ZONE; OLDER THAN **5 MINUTES** IS REFUSED.
- `used_percent_by_window`: PERCENT **USED** (0 TO 100) OF EACH USAGE LIMIT THE ASSISTANT HAS (FOR EXAMPLE ITS SESSION AND WEEKLY LIMITS). IF ANY IS AT THE RESERVE, THE WRITE IS REFUSED: **75%** USED, OR **90%** FOR `"reporter": "claude"`.
- `reporter`: A SHORT NAME OF THE ASSISTANT; `ordinary_usage_allowed`: MUST BE `true`.

WRITES YOU START YOURSELF IN THE WINDOW NEVER NEED THIS FILE.

6. **CHECK WHAT IT SAYS.** AN ASSISTANT CAN BE WRONG. READ THE APPROVE WINDOW YOURSELF: IT NAMES THE EXACT IMAGE THAT WILL BE WRITTEN.

CONNECT AN ASSISTANT (MCP)

CHROMATIC MANAGER HAS AN MCP SERVER (`scripts/chromatic_manager_mcp.py`). IT ATTACHES TO THE OPEN WINDOW THROUGH A LOCAL, TOKEN-PROTECTED API ON `127.0.0.1`.

CLAUDE CODE:

```
claude mcp add chromatic-manager -- "$env:LOCALAPPDATA\Jellymatic\venv\Scripts\python.exe"
-B "C:\path\to\Jellymatic\scripts\chromatic_manager_mcp.py"
```

CODEX (`~/.codex/config.toml`):

```
[mcp_servers.chromatic-manager]
# the full path of %LOCALAPPDATA%\Jellymatic\venv\Scripts\python.exe on your PC
command = 'C:\path\to\LocalAppData\Jellymatic\venv\Scripts\python.exe'
args = ['-B', 'C:\path\to\Jellymatic\scripts\chromatic_manager_mcp.py']
```

OTHER MCP CLIENTS: THE SAME COMMAND AND ARGUMENTS, AS A STDIO SERVER.

PROMPTS THAT ONLY READ

> USE THE CHROMATIC-MANAGER TOOLS TO CHECK MY CHROMATIC AND TELL ME, IN PLAIN WORDS, WHAT IS INSTALLED AND WHETHER ANYTHING NEEDS DOING. DO NOT START ANY JOB.

LIST MY CHROMATIC BACKUPS (PATHS ONLY) AND TELL ME WHETHER BOTH COPIES EXIST.

READ THE LAST CHROMATIC MANAGER JOB'S LOG AND EXPLAIN WHAT HAPPENED. DO NOT CHANGE ANYTHING.

PROMPTS THAT MAY CHANGE THE CONSOLE

MY CHROMATIC IS PLUGGED IN, ON, AND I AM HERE. ASK CHROMATIC MANAGER TO INSTALL THE START-UP MENU. I WILL APPROVE IT IN THE WINDOW. STOP AND TELL ME IF IT ASKS FOR ANYTHING ELSE.

ASK CHROMATIC MANAGER TO RUN "UPDATE THE POCKET LINK". I WILL CHECK A GAME ON THE CONSOLE WHILE THE NEW IMAGE RUNS FROM RAM AND APPROVE IT MYSELF. IF ANY AUTOMATIC CHECK FAILS, STOP AND EXPLAIN.

SOMETHING WENT WRONG DURING AN INSTALL. ASK CHROMATIC MANAGER TO FINISH THE INTERRUPTED JOB, AND TELL ME WHAT IT SAYS. DO NOT START ANYTHING ELSE.

PROMPTS FOR DEVELOPERS

IN THIS REPOSITORY, RUN THE HOST TESTS (SCRIPTS/TEST_CHROMATIC_LAUNCHER.PY, TEST_CHROMATIC_SETUP.PY, TEST_CHROMATIC_MANAGER.PY) AND SUMMARISE FAILURES. DO NOT OPEN ANY DEVICE OR SERIAL PORT.

EXPLAIN HOW THE START-UP MENU DECIDES BETWEEN THE MENU AND THE OFFICIAL FIRMWARE (CHROMATIC/LAUNCHER/BOOTLOADER_COMPONENTS, SHARED/BOOT_DECISION.H). DO NOT CHANGE CODE.

ADD A TEST TO SCRIPTS/TEST_CHROMATIC_SETUP.PY FOR <BEHAVIOUR>. KEEP THE POWER-CUT MODEL PASSING.

PROMPTS FOR BUG REPORTS

> HELP ME WRITE A BUG REPORT FOR JELLYMATIC FROM THIS CHROMATIC MANAGER LOG. REMOVE EVERY ADDRESS, NETWORK NAME, USER NAME, TOKEN AND DEVICE ID BEFORE SHOWING IT TO ME.

NEVER PASTE LOGS THAT SHOW ADDRESSES OR NAMES INTO A PUBLIC ISSUE. SEE SECURITY.MD.

CREDITS & LICENCES

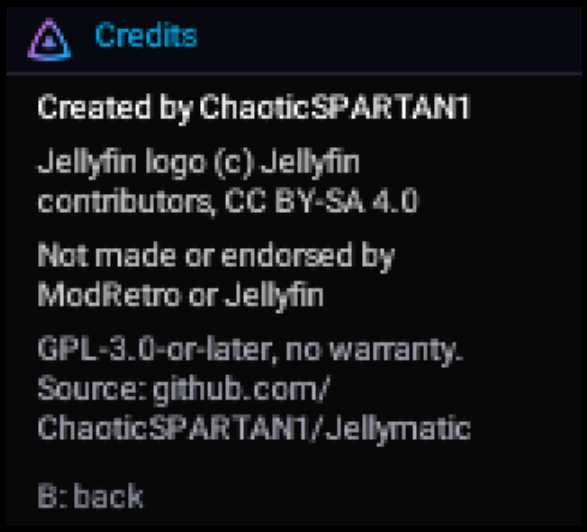
Credits & licences

HOME › CREDITS & LICENCES

CONTENTS: AUTHOR · THIS PROJECT'S LICENCE · PARTS MADE BY OTHERS · USED, NOT INCLUDED · READ AS REFERENCES · TRADEMARKS

AUTHOR

CREATED BY **CHAOTICSPARTAN1**. UNDER GPLV3 SECTION 7(B), REDISTRIBUTIONS AND MODIFIED VERSIONS MUST KEEP THIS ATTRIBUTION IN THE README CREDITS AND IN ANY ABOUT/CREDITS SCREEN (ADDITIONAL-TERMS.MD). THE START-UP MENU'S CREDITS SCREEN AND CHROMATIC MANAGER'S ABOUT TAB SHOW IT.



THIS PROJECT'S LICENCE

GPL-3.0-OR-LATER (LICENSE), FOR EVERYTHING WRITTEN FOR JELLYMATIC: THE START-UP MENU, THE JELLYFIN APP, THE RTL IN `rtl/` AND `chromatic/fpga-media/`, CHROMATIC MANAGER, THE SCRIPTS, REMOTE PILOT AND THE DOCUMENTATION. ADDITIONAL-TERMS.MD ADDS THE ATTRIBUTION TERM (7(B)), THE MARKING OF MODIFIED VERSIONS (7(C)), AND TWO ADDITIONAL PERMISSIONS: FOR FPGA BITSTREAMS THAT COMBINE JELLYMATIC'S RTL WITH MODRETRO'S DESIGN AND GOWIN IP, AND FOR LINKING THE CONSOLE FIRMWARE WITH ESPRESSIF'S BINARY-ONLY WI-FI/PHY LIBRARIES AND THE ESP32 ROM.

PARTS MADE BY OTHERS

PART	WHERE	LICENCE	NOTES
MODRETRO CHROMATIC FPGA DESIGN V18.14, FILES MODIFIED BY CHAOTICSPARTAN1 FOR JELLYMATIC, 2026-11	`fpga/images/*/src/` (THE LIST IS IN NOTICE)	GPL-3.0	© MODRETRO AND CONTRIBUTORS; THE GAME CORE FILES (`gb.v` AND OTHERS) COME FROM THE MISTER HANDHELD CORE "GAMEBOY_MISTER" (GPL-3.0-OR-LATER, TILL HARBAUM AND THE MISTER DEVELOPERS). `uart.v` NAMES GOWIN (GAOYUN SEMI) AS ITS CREATOR; MODRETRO PUBLISHES IT UNDER ITS REPOSITORY'S GPL. ONLY THE CHANGED FILES ARE HERE; THE REST IS MODRETRO'S PUBLIC REPOSITORY AT THE COMMIT IN `upstream.lock.json`.

PART	WHERE	LICENCE	NOTES
SHINY RELAY 1.1.5 PARTS	THE `uart_tx_busy` FIX IN THE 18.41 `system_monitor.sv`; THE REQUEST-ONCE STATUS CHANNEL PATTERN IN `rtl/chromatic_pocket_link.v`; THE WI-FI JOIN WORDING IN `chromatic/launcher/shared/wifi_scan.h`	GPL-3.0	© SHINY GEN. MARKED IN EACH FILE.
`speedcontrol.vhd`, VERILOG TRANSLATION FOR SIMULATION	`chromatic/fpga-media/sim/speedcontrol_model.v`	GPL-3.0 (AS THE ORIGINAL; NOT UNDER ADDITIONAL-TERMS.MD)	FROM MODRETRO'S DESIGN, WHICH HAS IT FROM THE MISTER CORE "GAMEBOY_MISTER"; TRANSLATED BY CHAOTICSPARTAN1.
"SIMPLE UART FOR FPGA", VERILOG PORT FOR SIMULATION	`rtl/uart2_vhdl_port.v`	MIT (THE PORT: GPL-3.0-OR-LATER)	© JAKUB CABAL; THE VHDL ORIGINAL IS PART OF MODRETRO'S DESIGN. LICENSES/MIT-UART_FOR_FPGA.TXT
ESP-IDF BOOTLOADER START FILE	`chromatic/launcher/bootloader_components/main/bootloader_start.c`	APACHE-2.0	© ESPRESSIF SYSTEMS; MODIFIED (THE START DECISION). LICENSES/APACHE-2.0.TXT

PART	WHERE	LICENCE	NOTES
JELLYFIN LOGO, SCALED	`chromatic/launcher/shared/launcher_art.h`, `scripts/chromatic_theme.py` (JELLYFIN_MARK), PICTURES IN `docs/images/` SHOWING IT	CC BY-SA 4.0	© THE JELLYFIN CONTRIBUTORS, FROM JELLYFIN/JELLYFIN-UX. THE SCALED VERSIONS ARE SHARED UNDER THE SAME LICENCE. LICENSES/CC-BY-SA-4.0-JELLYFIN-LOGO.TXT
ROBOTO MEDIUM, RASTERIZED	`pocket/ui_font.h` (MADE BY `scripts/build_ui_font.py`)	APACHE-2.0	ROBOTO 2.138 BY CHRISTIAN ROBERTSON, COPYRIGHT 2011 GOOGLE INC. LICENSES/APACHE-2.0.TXT
T80 CPU CORE, SAMEBOY-DERIVED BOOT ROM	NOT HERE: IN MODRETRO'S SOURCE, BUILT INTO THE FPGA IMAGES 18.40 AND 18.41	BSD-STYLE; MIT	© DANIEL WALLNER; © LIOR HALPHON. THEIR NOTICES GO WITH THE IMAGES: LICENSES/BSD-STYLE-T80.TXT, LICENSES/MIT-SAMEBOY-BOOT-ROM.TXT
PRESS START 2P FONT	`scripts/chromatic_gui_fonts/`, `docs/fonts/`, `remote-pilot/fonts/`	SIL OFL 1.1	CODEMAN38. LICENSES/OFL-1.1-PRESSSTART2P.TXT
SILKSCREEN FONT	`scripts/chromatic_gui_fonts/`, `docs/fonts/`, `remote-pilot/fonts/`	SIL OFL 1.1	THE SILKSCREEN PROJECT AUTHORS. LICENSES/OFL-1.1-SILKSCREEN.TXT

USED, NOT INCLUDED

INSTALLED BY YOU, UNDER THEIR OWN LICENCES:

TOOL	LICENCE
ESP-IDF V5.3 (ESPRESSIF) AND ITS COMPONENTS (FREERTOS, LWIP, MBED TLS, NEWLIB, WI-FI LIBRARIES)	APACHE-2.0 AND OTHERS; SEE ESP-IDF'S `COPYRIGHT.rst`. A **BUILT** START-UP MENU CONTAINS THESE AND MUST SHIP THEIR NOTICES.
MODRETRO'S FPGA SOURCE (WITH THE MISTER CORE ("GAMEBOY_MISTER"), T80 CPU, SAMEBOY-DERIVED BOOT ROM, GOWIN IP)	GPL-3.0 / GPL-3.0-OR-LATER / BSD-STYLE / MIT / GOWIN'S TERMS. A **BUILT** FPGA IMAGE CONTAINS THESE AND MUST SHIP THEIR NOTICES AND SOURCE.
GOWIN EDA	GOWIN'S LICENCE (PROPRIETARY). NEVER INCLUDED.
OPENFPGALOADER	APACHE-2.0
PYTHON, PYSIDE6 / QT	PSF LICENCE; LGPL-3.0
ESPTOOL, PYSERIAL	GPL-2.0-OR-LATER; BSD-3-CLAUSE
MODRETRO'S OFFICIAL UPDATE TOOL	MODRETRO'S TERMS. NEVER INCLUDED; DOWNLOAD IT FROM MODRETRO.

READ AS REFERENCES

THESE WERE READ TO UNDERSTAND THE CONSOLE; NO CODE WAS COPIED FROM THEM EXCEPT THE PARTS LISTED ABOVE: MODRETRO'S CHROMATIC MCU SOURCE (GPL-3.0), **CHROMATIX** BY ENJOY-DIGITAL AND CONTRIBUTORS (GPL-3.0 / BSD-2-CLAUSE), AND **SHINY RELAY** BY SHINY GEN (GPL-3.0; BESIDES THE PARTS LISTED ABOVE, THE "DEBUG BIT" VERSION SCHEME FOLLOWS ITS APPROACH). THANK YOU TO ALL OF THEM.

TRADEMARKS

MODRETRO AND CHROMATIC ARE TRADEMARKS OF MODRETRO, INC. JELLYFIN IS A TRADEMARK OF ITS OWNERS. ANALOGUE AND POCKET ARE TRADEMARKS OF ANALOGUE. THEY ARE USED ONLY TO SAY WHAT JELLYMATIC WORKS WITH. JELLYMATIC IS AN INDEPENDENT PROJECT, NOT AFFILIATED WITH, ENDORSED OR SUPPORTED BY ANY OF THEM.

Design notes (for developers)

SOME COMMENTS POINT TO "DOCS/DESIGN-NOTES.MD", THIS PAGE. A FEW, IN THE REVIEWED FPGA SOURCES (`fpga/images/`, THE PINNED FILES IN `rtl/` AND `chromatic/fpga-media/`) AND IN THE STAGING SCRIPTS WHOSE TEXT BECOMES PART OF THOSE SOURCES, STILL NAME THE ORIGINAL INTERNAL DESIGN NOTES (`CHROMATIC-...md`). THOSE NOTES ARE NOT PUBLISHED, AND THE PINNED SOURCES CANNOT BE EDITED WITHOUT CHANGING THEIR SHA-256 HASHES. THIS PAGE SAYS WHERE EACH TOPIC LIVES NOW.

WHAT THEY COVERED, AND WHERE TO LOOK INSTEAD:

NOTE	TOPIC	WHERE NOW
POCKET-TETHER	CHROMATIC ⇔ POCKET WIRE CONTRACT, CREDITS, MEMORY BUDGET, LINK STAGES	`shared/pocket_tether.h`, `chromatic/launcher/shared/pocket_link.h`, `rtl/chromatic_pocket_link.v` AND THEIR TESTS; POCKET LINK
REMOTE-CONTROL	CHR1 PROTOCOL (REMOTE PILOT ⇔ LAUNCHER), USB PACING, PAIRING	`chromatic/launcher/shared/remote.h`, `remote-pilot/presets/chromatic/PROTOCOL.md`; REMOTE PILOT
AUDIO-PLAN	FPGA SOUND HOOKS, VERSION REGISTER, THE FPGA JOB RUNBOOK	`chromatic/fpga-media/chromatic_media.v`, `scripts/stage_chromatic_fpga_media.py`, `scripts/chromatic_fpga.py`; INSTALL, RECOVERY
SETUP-ON-CONSOLE	WI-FI AND JELLYFIN SCREENS, THE ONE AUDITED FLASH WRITER, POWER-CUT MODEL	`chromatic/launcher/shared/setup_ui.h`, `provision_write.c`, `scripts/test_chromatic_setup.py`; FIRST RUN
JELLYFIN-PLAYER	THE CONSOLE'S JELLYFIN APP, PACING, MEMORY	`chromatic/jellyfin/jf_app.h`, `jf_ui.h`; USING JELLYFIN

NOTE	TOPIC	WHERE NOW
FPGA-STREAM-IMAGE	HOW THE 18.41 IMAGE MET TIMING, ITS PINS	`fpga/images/link-18.41/` RECEIPTS, `scripts/stage_chromatic_fpga_link.py`
DIAGNOSTICS	THE LAUNCHER'S `PJ-DIAG` SERIAL LINES	`chromatic/launcher/shared/launcher_diag.h`
MANAGER	CHROMATIC MANAGER'S BEHAVIOUR IN DETAIL	`scripts/chromatic_manager*.py` DOCSTRINGS AND TESTS; INSTALL

THE PROVISIONING RECORD (WI-FI, PAIRING, JELLYFIN SIGN-IN IN TWO FLASH SECTORS) IS DEFINED BY `scripts/chromatic_provision.py` AND ITS C TWIN `chromatic/launcher/shared/provision.h`, CROSS-CHECKED BY `scripts/test_chromatic_provision_c.py`.

CONTRIBUTING



Contributing to Jellymatic

PULL REQUESTS, BUG FIXES AND BUG REPORTS ARE WELCOME. THIS IS A HOBBY PROJECT: THERE IS NO GUARANTEE OF A RESPONSE OR A FIX, BUT EVERY REPORT IS READ WHEN TIME ALLOWS.

REPORTING A BUG

OPEN AN ISSUE WITH:

1. WHAT YOU DID, WHAT YOU EXPECTED, WHAT HAPPENED.
2. VERSIONS: START-UP MENU (SHOWN ON ITS CREDITS SCREEN AND IN CHROMATIC MANAGER), CHROMATIC MANAGER COMMIT, FPGA IMAGE (OFFICIAL, 18.40 OR 18.41), WINDOWS VERSION.
3. THE MANAGER'S RESULT HEADLINE AND, IF USEFUL, THE RELEVANT LOG LINES **AFTER REMOVING** EVERY ADDRESS, NETWORK NAME, USER NAME, TOKEN AND DEVICE ID. TURN ON THE MANAGER'S EYE BUTTON (PRIVATE DETAILS HIDDEN) BEFORE TAKING SCREENSHOTS.

NEVER POST WI-FI DETAILS, SERVER ADDRESSES, JELLYFIN TOKENS, BACKUPS OR FULL LOGS. FOR SECURITY PROBLEMS USE SECURITY.MD INSTEAD OF AN ISSUE.

PULL REQUESTS

- KEEP THE SAFETY RULES: NO CHANGE MAY WRITE THE OFFICIAL FIRMWARE SLOT, SETTINGS OR CARTRIDGES; EVERY WRITE GOES THROUGH CHROMATIC MANAGER'S JOURNAL, BACKUP, READ-BACK AND APPROVE.
- RUN THE TESTS THAT COVER YOUR CHANGE (DOCS/BUILDING-FROM-SOURCE.MD) AND SAY WHICH ONES YOU RAN. THE TESTS MUST NEVER OPEN A DEVICE.
- PLAIN WORDS IN ANYTHING A USER READS. KEEP THE PIXEL STYLE OF THE CONSOLE AND THE MANAGER.
- AFTER EDITING ANY DOCUMENTATION PAGE, RUN `py -3 scripts\build_docs_html.py` (OPEN-DOCUMENTATION.HTML AND DOCS/INDEX.HTML) AND `py -3 scripts\build_docs_pdf.py` (THE PDF GUIDE); `scripts\test_docs_html.py` CHECKS ALL

THREE. WHEN THE REPOSITORY IS PUBLIC, GITHUB PAGES CAN SERVE THE DOCS: SETTINGS > PAGES, BRANCH `main`, FOLDER `/docs`; THE SITE IS `https://chaoticspartan1.github.io/Jellymatic/`.

- DO NOT ADD FILES YOU MAY NOT SHARE (FIRMWARE DUMPS, MODRETRO'S OR GOWIN'S FILES, LICENCES, REAL MEDIA).
- BY CONTRIBUTING YOU AGREE THAT YOUR CONTRIBUTION IS LICENSED UNDER GPL-3.0-OR-LATER WITH THE ADDITIONAL ATTRIBUTION TERM IN ADDITIONAL-TERMS.MD.

SECURITY



JELLYMATIC / SECURITY.MD

Security

REPORTING A VULNERABILITY

PLEASE REPORT SECURITY PROBLEMS **PRIVATELY**, NOT IN A PUBLIC ISSUE: USE GITHUB'S **SECURITY › REPORT A VULNERABILITY** (PRIVATE VULNERABILITY REPORTING) ON THIS REPOSITORY. DESCRIBE THE PROBLEM, HOW TO REPRODUCE IT, AND WHAT IT AFFECTS (CONSOLE, PC TOOL, REMOTE PILOT, JELLYFIN SIGN-IN). THERE IS NO GUARANTEE OF A RESPONSE TIME; REPORTS ARE HANDLED AS TIME ALLOWS.

NEVER POST PUBLICLY

- WI-FI NETWORK NAMES AND PASSWORDS
- JELLYFIN SERVER ADDRESSES, USER NAMES, USER IDS, TOKENS, QUICK CONNECT CODES, DEVICE IDS
- REMOTE PILOT PAIRING KEYS, THE MANAGER'S LOCAL API TOKEN OR DISCOVERY FILE
- CONSOLE BACKUPS, FLASH DUMPS, SERIAL NUMBERS, MAC ADDRESSES
- LOGS OR SCREENSHOTS THAT SHOW ANY OF THE ABOVE (TURN ON THE MANAGER'S EYE BUTTON BEFORE A SCREENSHOT)

WHAT JELLYMATIC DOES TO PROTECT YOU

- YOUR WI-FI AND JELLYFIN DETAILS STAY IN `%LOCALAPPDATA%\Jellymatic` ON YOUR PC AND ON YOUR CONSOLE; NOTHING IS UPLOADED.
- THE CONSOLE GETS ITS OWN REVOCABLE JELLYFIN SIGN-IN (QUICK CONNECT); YOUR PASSWORD NEVER GOES TO IT, AND JELLYMATIC SIGNS OUT ONLY THAT CONSOLE'S OWN SIGN-IN.
- CHROMATIC MANAGER'S API LISTENS ON 127.0.0.1 ONLY, WITH A RANDOM TOKEN PER RUN; EVERY WRITE NEEDS YOUR APPROVE IN THE WINDOW.
- REMOTE PILOT WORKS ON PRIVATE NETWORKS ONLY AND NEVER SHOWS PASSWORD FIELDS WHILE CONNECTED.

THIRD-PARTY NOTICES



JELLYMATIC / THIRD-PARTY-NOTICES.MD

Third-party notices

COMPONENTS BY OTHERS THAT ARE **INCLUDED IN THIS REPOSITORY**, WITH WHERE THEIR LICENCE TEXT IS.

COMPONENT	FILES HERE	LICENCE	LICENCE TEXT
MODRETRO CHROMATIC FPGA DESIGN V18.14 (COMMIT A01AA375), FILES MODIFIED BY CHAOTICSPARTAN1 FOR JELLYMATIC, 2026-11	`fpga/images/sound-18.40/src/` AND `fpga/images/link-18.41/src/`: MODRETRO'S FILES `top.v`, `rtl/BSP/system_monitor.sv`, `rtl/EMU/emu_system_top.v`, `rtl/EMU/cart.v`, `board/evt1_x2/evt1_x2.sdc` (BOTH IMAGES), `rtl/BSP/uart/uart.v`, AND THE PROJECT FILES `evt1_x2.gprj`, `build_options.tcl` (18.41); THE FULL LIST IS IN NOTICE	GPL-3.0	LICENSE (THE SAME GPL-3.0 TEXT)

COMPONENT	FILES HERE	LICENCE	LICENCE TEXT
`uart.v` (UART2 WRAPPER) IN MODRETRO'S DESIGN	`fpga/images/link-18.41/src/rtl/BSP/uart/uart.v` (MODIFIED)	GPL-3.0, AS MODRETRO PUBLISHES IT IN ITS REPOSITORY	LICENSE. THE FILE CAME TO MODRETRO'S DESIGN FROM GOWIN'S REFERENCE DESIGN; MODRETRO PUBLISHES IT UNDER ITS REPOSITORY'S GPL. KEPT AS MODRETRO SHIPS IT APART FROM JELLYMATIC'S MARKED HOOK.
MISTER CORE ("GAMEBOY_MISTER") (TILL HARBAUM AND THE MISTER DEVELOPERS), AS USED IN MODRETRO'S DESIGN	`fpga/images/*/src/rtl/EMU/CORE/gb.v` (MODIFIED)	GPL-3.0-OR-LATER (FILE HEADER)	LICENSE

COMPONENT	FILES HERE	LICENCE	LICENCE TEXT
SHINY RELAY 1.1.5 (SHINY GEN), THE OPEN FIRMWARE FOR THE CHROMATIC	THE `uart_tx_busy` ONE-CYCLE GAP FIX IN `fpga/images/link-18.41/src/rtl/BSP/system_monitor.sv` (TAKEN FROM ITS `system_monitor.sv`, MARKED IN THE FILE); THE REQUEST-ONCE STATUS CHANNEL PATTERN OF ITS `link_ev_req` IN `rtl/chromatic_pocket_link.v`; THE PLAIN-WORD WI-FI JOIN MESSAGES IN `chromatic/launcher/shared/wifi_scan.h`	GPL-3.0	LICENSE (THE SAME GPL-3.0 TEXT)
`speedcontrol.vhd` OF MODRETRO'S DESIGN (FROM THE MISTER CORE "GAMEBOY_MISTER"), TRANSLATED TO VERILOG FOR SIMULATION	`chromatic/fpga-media/sim/speedcontrol_model.v` (TRANSLATION BY CHAOTICSPARTAN1, 2026-11)	GPL-3.0, AS MODRETRO PUBLISHES THE ORIGINAL (ADDITIONAL-TERMS.MD DOES NOT APPLY)	LICENSE
"SIMPLE UART FOR FPGA" (JAKUB CABAL), AS USED IN MODRETRO'S DESIGN; VERILOG PORT FOR SIMULATION	`rtl/uart2_vhdl_port.v`	MIT (PORT: GPL-3.0-OR-LATER)	LICENSES/MIT-UART_FOR_FPGA.TXT

COMPONENT	FILES HERE	LICENCE	LICENCE TEXT
ESPRESSIF ESP-IDF BOOTLOADER START FILE	`chromatic/launcher/bootloader_components/main/bootloader_start.c` (MODIFIED)	APACHE-2.0	LICENSES/APACHE-2.0.TXT
JELLYFIN LOGO (JELLYFIN/JELLYFIN-UX, COMMIT 69A2B7B1)	SCALED IN `chromatic/launcher/shared/launcher_art.h`, `scripts/chromatic_theme.py`; SHOWN IN `docs/images/jellymatic-icon.png` AND CONSOLE SCREENSHOTS	CC BY-SA 4.0	LICENSES/CC-BY-SA-4.0-JELLYFIN-LOGO.TXT
ROBOTO MEDIUM (ROBOTO 2.138, COPYRIGHT 2011 GOOGLE INC.), RASTERIZED AND SCALED BY `scripts/build_ui_font.py`	`pocket/ui_font.h`	APACHE-2.0	LICENSES/APACHE-2.0.TXT
PRESS START 2P (CODEMAN38)	`scripts/chromatic_gui_fonts/heading-PressStart2P-Regular.ttf`, `docs/fonts/`, `remote-pilot/fonts/`	SIL OFL 1.1	LICENSES/OFL-1.1-PRESSSTART2P.TXT
SILKSCREEN (THE SILKSCREEN PROJECT AUTHORS)	`scripts/chromatic_gui_fonts/body-Silkscreen-Regular.ttf`, `docs/fonts/`, `remote-pilot/fonts/`	SIL OFL 1.1	LICENSES/OFL-1.1-SILKSCREEN.TXT

NOT INCLUDED, NEEDED TO BUILD OR RUN

INSTALLED SEPARATELY BY YOU; EACH COMES WITH ITS OWN LICENCE: ESP-IDF V5.3 AND ITS COMPONENTS, MODRETRO'S FULL FPGA SOURCE WITH ITS SUBMODULE, GOWIN EDA (PROPRIETARY, NEEDS YOUR OWN LICENCE), OPENFPGALOADER (APACHE-2.0), PYTHON (PSF), PYSIDE6/QT (LGPL-3.0), ESPTOOL (GPL-2.0-OR-LATER), PYSERIAL (BSD-3-CLAUSE).

IF YOU DISTRIBUTE BUILT IMAGES

THIS REPOSITORY SHIPS **NO** BUILT FIRMWARE. IF YOU PUBLISH A BUILT START-UP MENU OR FPGA IMAGE:

- **START-UP MENU** (ESP32 BINARIES): INCLUDE ESP-IDF'S NOTICES (ITS `LICENSE` AND `COPYRIGHT.rst`, WHICH LIST FREERTOS (MIT), LWIP (BSD-3-CLAUSE), MBED TLS (APACHE-2.0), WPA_SUPPLICANT (BSD), NEWLIB AND THE ESPRESSIF WI-FI/PHY LIBRARIES (APACHE-2.0)), THIS REPOSITORY'S LICENCE, AND THE EXACT SOURCE COMMIT. LINKING WITH ESPRESSIF'S BINARY-ONLY WI-FI/PHY LIBRARIES AND CALLING THE ESP32 ROM IS ALLOWED BY THE ADDITIONAL PERMISSION IN ADDITIONAL-TERMS.MD, SECTION 4.
- **FPGA IMAGES**: THEY ARE GPL WORKS. SHIP THEM WITH THE GPL TEXT, THIS REPOSITORY'S SOURCE AT THE SAME COMMIT AND MODRETRO'S SOURCE AT THE COMMIT IN `upstream.lock.json`, PLUS THE NOTICES OF THE T80 CPU (LICENSES/BSD-STYLE-T80.TXT, REQUIRED FOR SYNTHESIZED FORMS) AND THE SAMEBOY-DERIVED BOOT ROM (LICENSES/MIT-SAMEBOY-BOOT-ROM.TXT). BOTH ARE IN MODRETRO'S SOURCE, NOT IN THIS REPOSITORY, BUT BOTH ARE BUILT INTO THE IMAGES. THE IMAGES ALSO CONTAIN GOWIN IP THAT MODRETRO PUBLISHES ONLY IN ENCRYPTED FORM; CHECK GOWIN'S TERMS FOR REDISTRIBUTING BITSTREAMS BEFORE YOU PUBLISH ONE.

ADDITIONAL TERMS



JELLYMATIC / ADDITIONAL-TERMS.MD

Additional terms (GNU GPL version 3, section 7)

JELLYMATIC IS LICENSED UNDER THE GNU GENERAL PUBLIC LICENSE, VERSION 3 OR (AT YOUR OPTION) ANY LATER VERSION (LICENSE). THESE ADDITIONAL TERMS APPLY ONLY TO THE MATERIAL IN THIS REPOSITORY WHOSE COPYRIGHT IS HELD BY ITS AUTHOR, **CHAOTICSPARTAN1**. THEY DO NOT APPLY TO PARTS MADE BY OTHERS (SEE NOTICE AND THIRD-PARTY-NOTICES.MD), WHICH KEEP THEIR OWN LICENCES, NOR TO TRANSLATIONS OF OTHERS' FILES SUCH AS `chromatic/fpga-media/sim/speedcontrol_model.v` (GPL-3.0, AS ITS ORIGINAL).

1. ATTRIBUTION (SECTION 7(B))

AS PERMITTED BY GPLV3 SECTION 7(B), IF YOU CONVEY THIS WORK OR A MODIFIED VERSION OF IT, YOU MUST PRESERVE THE REASONABLE AUTHOR ATTRIBUTION "CHAOTICSPARTAN1":

1. IN THE CREDITS SECTION OF THE README (OR THE EQUIVALENT MAIN DOCUMENTATION PAGE), AND
2. IN ANY ABOUT, CREDITS OR SIMILAR SCREEN THAT THE PROGRAM OR ITS USER INTERFACE SHOWS (FOR EXAMPLE THE START-UP MENU'S CREDITS SCREEN AND CHROMATIC MANAGER'S ABOUT TAB).

2. MARKING MODIFIED VERSIONS (SECTION 7(C))

AS PERMITTED BY GPLV3 SECTION 7(C), A MODIFIED VERSION MUST BE MARKED IN REASONABLE WAYS AS DIFFERENT FROM THE ORIGINAL VERSION (FOR EXAMPLE IN ITS NAME, VERSION STRING OR ABOUT/CREDITS SCREEN), AND MUST NOT BE PRESENTED AS THE ORIGINAL JELLYMATIC OR AS MADE OR ENDORSED BY ITS AUTHOR.

3. ADDITIONAL PERMISSION FOR FPGA BITSTREAMS (SECTION 7, ADDITIONAL PERMISSION)

IF YOU MODIFY THIS PROGRAM, OR ANY COVERED WORK, BY COMBINING OR LINKING IT WITH MODRETRO'S CHROMATIC FPGA DESIGN (OR A MODIFIED VERSION OF IT) OR WITH GOWIN SEMICONDUCTOR FPGA IP OR PRIMITIVES GENERATED OR SUPPLIED BY GOWIN'S FPGA DESIGN SOFTWARE (OR MODIFIED

VERSIONS OF THEM), THE LICENSORS OF THIS PROGRAM GRANT YOU ADDITIONAL PERMISSION TO CONVEY THE RESULTING WORK, INCLUDING FPGA BITSTREAMS. CORRESPONDING SOURCE FOR A NON-SOURCE FORM OF SUCH A COMBINATION NEED NOT INCLUDE THE SOURCE CODE OF THOSE PARTS, WHICH REMAIN UNDER THEIR OWN LICENCES.

THIS PERMISSION IS GIVEN ONLY FOR THE COPYRIGHT HELD BY CHAOTICSPARTAN1 (THE RTL IN `rtl/` AND `chromatic/fpga-media/`, AND THE BUILD SCRIPTS). IT CANNOT AND DOES NOT CHANGE THE LICENCE OF MODRETRO'S DESIGN, WHICH IS ITSELF GPL-3.0: ANYONE CONVEYING A BITSTREAM MUST STILL MEET MODRETRO'S LICENCE TERMS AND GOWIN'S TERMS FOR ITS IP. SEE THIRD-PARTY-NOTICES.MD.

4. ADDITIONAL PERMISSION FOR ESPRESSIF'S BINARY-ONLY LIBRARIES AND THE ESP32 ROM (SECTION 7, ADDITIONAL PERMISSION)

IF YOU MODIFY THIS PROGRAM, OR ANY COVERED WORK, BY LINKING OR COMBINING IT WITH THE BINARY-ONLY LIBRARIES THAT ESPRESSIF SYSTEMS SUPPLIES WITH ESP-IDF FOR THE ESP32 (SUCH AS THE WI-FI, PHY, BLUETOOTH, COEXISTENCE AND RF CALIBRATION LIBRARIES, FOR EXAMPLE `libnet80211`, `libpp`, `libphy`, `librtc` AND `libcoexist`, OR MODIFIED VERSIONS OF THEM THAT ESPRESSIF SUPPLIES), OR WITH THE CODE IN THE ESP32 CHIP'S MASK ROM, THE LICENSORS OF THIS PROGRAM GRANT YOU ADDITIONAL PERMISSION TO CONVEY THE RESULTING WORK. CORRESPONDING SOURCE FOR A NON-SOURCE FORM OF SUCH A COMBINATION NEED NOT INCLUDE THE SOURCE CODE OF THOSE LIBRARIES OR OF THE ROM, WHICH REMAIN UNDER ESPRESSIF'S OWN LICENCE TERMS; THE CORRESPONDING SOURCE OF THE PARTS COVERED BY THIS LICENCE MUST STILL BE PROVIDED.

THIS PERMISSION IS GIVEN ONLY FOR THE COPYRIGHT HELD BY CHAOTICSPARTAN1 (THE START-UP MENU, THE JELLYFIN APP AND THE OTHER CODE IN `chromatic/`, `shared/` AND `pocket/` THAT IS BUILT INTO THE CONSOLE'S ESP32 FIRMWARE). IT DOES NOT CHANGE THE LICENCE OF ANY OTHER PART.

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